

IV/PLAY USER GUIDE

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v.2.8.4 - Sunday, May 24, 2026

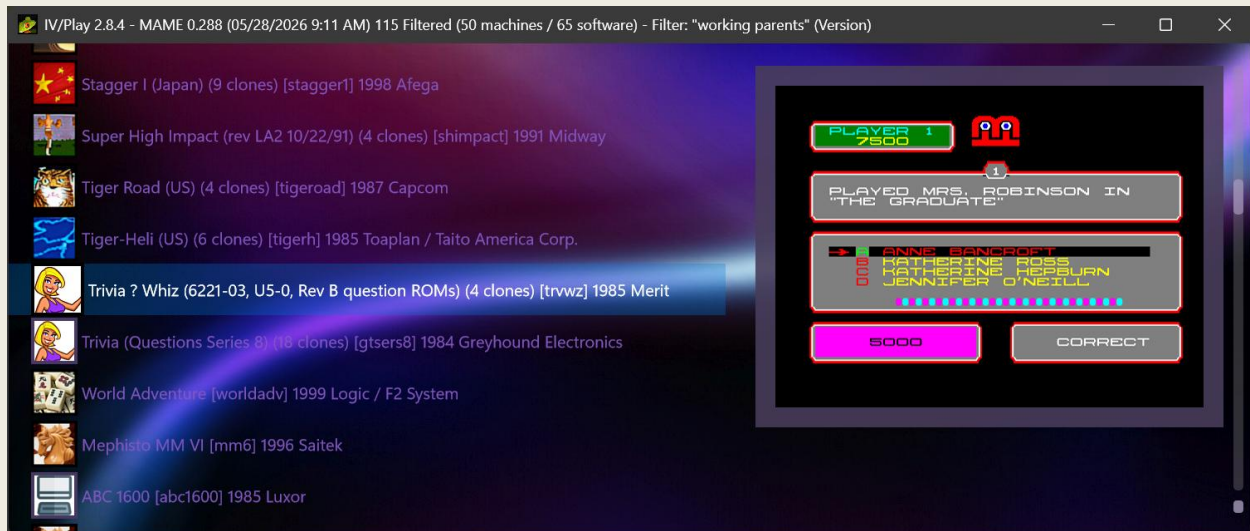


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OVERVIEW

IV/Play (pronounced 'Four Play') is a desktop/keyboard-oriented GUI front-end for [MAME™](#). It was designed and its coding commissioned in 2011 by original MAMEUI team member John IV, from his original 2006 concepts. It has a particular feature-set, is keyboard driven, and utilises many of the navigation and keyboard short cuts of MAMEUI. It is targeted towards the latest version of Microsoft Windows 11 (2026). IV/Play is decoupled from setting MAME options directly, using its (MAME's) MAME.ini in an effort to future proof and guard against core command line changes and updates. It is self-contained and portable for easy transport and mobility.

In 2025 a modernization was undertaken of the 14-year-old codebase to update from its .NET 4.5 WPFWrapper origins to its current state, rebuilt entirely using WinForms, with WPF fully deprecated. It provides a high performance, RAM based experience on i7-12700K, 64GB RAM, GPU level machines. The goal is near instant launch, game list population, and zero slowdown displaying any art assets (icons, snaps, flyers, bkgounds, etc.) utilizing as much RAM and graphic (GPU) hardware as needed; employing modern design techniques and technologies.

Naming conventions: The application name is 'IV/Play' but due to WinOS file-naming limitations, 'IV-Play' is used where necessary.

In the 1080p 96DPI era, the initial IV/Play dimensions were 1015x432; a 2.35:1 aspect ratio in honor of the 1960's [TohoScope](#) tech used in the filming of Godzilla movies. Today, on 4K monitors, the same ratio is used but at a default of 1522x648 (144dpi and 1.5X scaling).

IV/PLAY VS. MAMEUI

MAMEUI is an integrated UI to [MAME](#) and is compiled using the base source. It was the first 'official' MAME port to the Windows platform and the most popular version of MAME for decades. It leveraged the underlying MAME code and depended on it to produce its UI and functionality. This was its strength and weakness. As MAMEUI matured (b. 1997), the original development team waned over time. It often took long stretches to catch up with the sweeping core changes that periodically took place in the underlying MAME engine. While the number of supported arcade games, gambling 'fruit machines', computers, chess games, calculators, consoles, and miscellaneous other ROM based electronics neared 40,000 sets, the UI slowed down when all the icons and screenshots were processed by the near 3-decade old controls. This introduced memory leaks over time and the user experience degraded. Changes to the core at the time also resulted in huge memory footprints and 5x longer load times.

IV/Play was born out of the desire to continue a similar 'classic' desktop interface to MAMEUI without the attendant issues of being coupled to the core.

REQUIREMENTS

Windows 11 is the preferred platform. Faster modern processors and solid-state drives (SSDs / NVMe)s help considerably with image/resource loading speed and are strongly recommended. IV/Play 2.2.0 is fully DPI aware and will scale its UI and elements for a 4K monitor for example. IV/Play is built using .NET 10 via ATO and is included in the .exe so it will not prompt to download it from Microsoft. IV/Play, as mentioned in the overview is now (2025) being optimized for high end systems, which include the latest processors, copious amounts of RAM, and GPUs to drive its performant mission.

INSTALLATION

The most straightforward installation option is to place the IV/Play files into the top level of an existing MAME directory structure alongside the respective .exe. IV/Play will utilise the structure's icons, art types, and bkground folders of an old MAMEUI installation. IV/Play does not rely on registry settings or an install so it is easily portable on external media like a thumb drive with MAME and attendant files. IV/Play will create its owned named directory under its location for its various assets, but also maintain a same level favorites.ini file, custom game list and its *.cfg file, which allow them to survive the asset directory removal during testing or refresh. The attendant e_sqlite3.dll must also be present with the .exe.

RESOURCES

The current build, its recommended icons, and snapshots for IV/Play are found at the [IV/Play homepage](#).

MAJOR APPLICATION FUNCTIONS

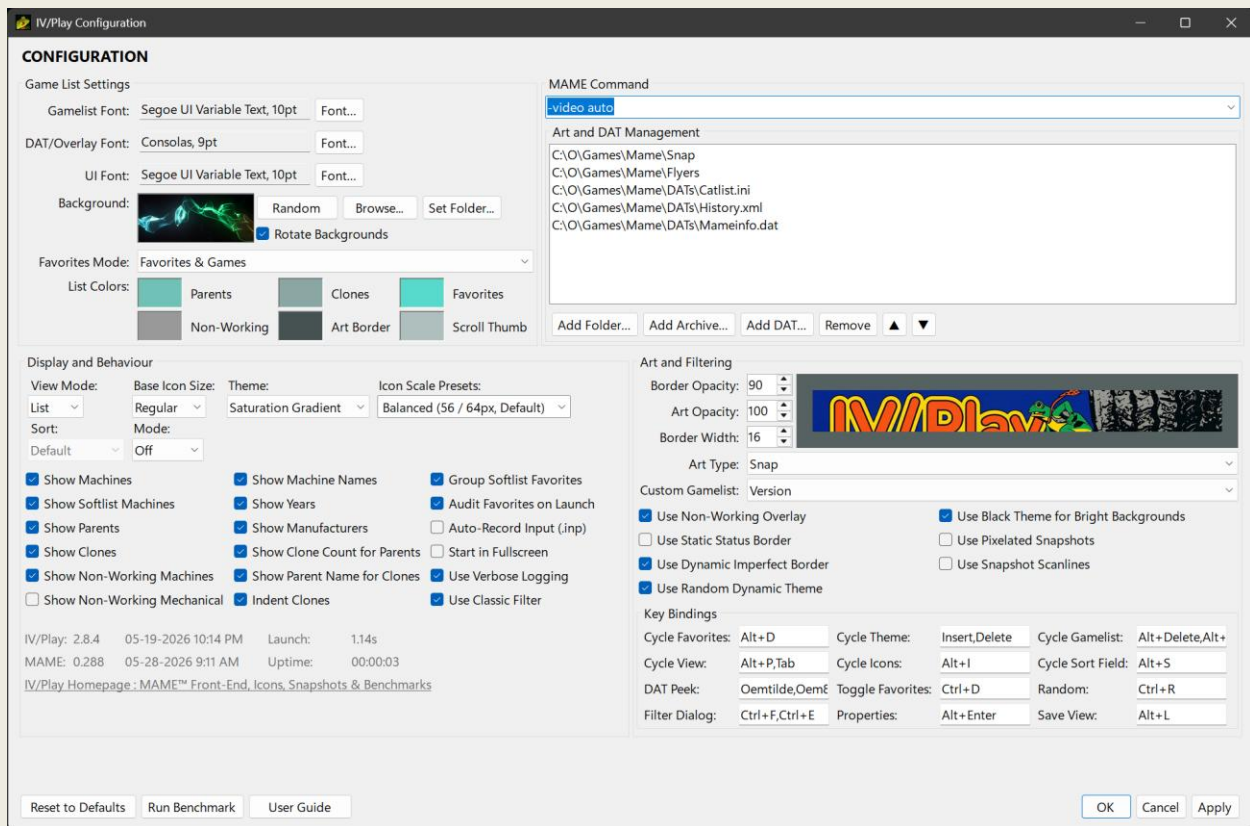
IV/Play is, at its core, a game launcher. It depends on the presence of a MAME.exe to import its XML (populating its list view and properties), and to pass back its game launch command. IV/Play is designed to be as low maintenance as possible and will not directly set MAME options; edit the baseline MAME.ini for that (though it is possible to use the **F1** command-line override to supersede the MAME.ini).

IMPORT XML DATA FROM MAME.EXE

IV/Play resides in the same directory as the MAME executable or in its own directory. Upon initial launch, it searches the local directory for a MAME version. Failing to find one a dialogue box appears asking to locate the MAME.exe to 'link'. It then grabs the output of the (currently) 296MB MAME XML file (MAME.exe -listxml). **F5** refreshes the XML using the same named MAME.exe, (when a new version of MAME is released for example) while **F4** re-opens the dialogue box to select a different MAMExx.exe.

CONFIGURATION SCREEN (F1)

F1 brings up the configuration screen. The UI settings are stored in the user editable IV-Play.cfg file, and may be viewed in-session with the **F3** key.



DISPLAY XML DATA IN UI

TITLE BAR

The title bar displays the version of MAME linked along with the number of games supported in that version and favorites in use. The number of favorites displayed will change if the favorites and filter are used.

GAME LIST

The game list displays all the games from the MAME XML output. Clones are automatically offset from their parents (but can have the indent removed in the **F1** screen). The font is selectable as is the color of the parents, clones, non-working, and favorites. Navigation through the list is via regular direction keys like **up**, **down**, **page up**, **page down**, **home** and **end**.

Display of the year, manufacturer, and machine (internal MAME) name is configurable through **F1**.

Free typing 'donkey kong' in the game list goes to the first instance depending where focus is when typing; if there are no instances downward it will go to the first instance of 'donkey kong' which may wrap around to the top in the favorites.

The Game List can be toggled with **Alt-P** and **TAB** to step through the Game List modes: regular icon list, large icon list, large icon grid, and large icon grid + no art.

FAVORITES

IV/Play supports the use of a `.\favorites.ini` file. This is a text file with one game per line, for example:

```
10yard
1941
dkong
frogger
```

Favorites are toggled by **Alt-D** and can be set in the **F1** configuration screen. The favorites are displayed above the MAME generated game list in alphabetical order. The icons are intentionally **not** offset for favorites to avoid confusion if a clone is added as a favorite and its parent is not there. The favorites can be displayed in different colors than the regular game list (set via **F1** in the configuration dialogue). The **Alt-D** toggle will cycle through favorites off, favorites + game list, favorites by themselves. Non-working and clone games added to favorites survive filters set on the lower game list.

Favorites are added by **Ctrl-D** from the game list and removed by **Ctrl-D** from the favorites area.

Up/Down/Left/Right/Page Up/Page Down behave the same way they do in the game list. A single press of **down arrow** on the last favorite goes to the first regular game list item. **Home** and **End** buttons on the keyboard provide a two-stop navigation system to move between the favorites and main game lists efficiently.

- **End Key:** When the focus is in the favorites list, the first press of **End** jumps to the last favorite. A second press jumps to the very end of the main game list. If the focus is already in the main list, it jumps directly to the end.
- **Home Key:** When the focus is in the main game list, the first press of **Home** jumps to the top of the main list (directly below favorites). A second press jumps to the top of the favorites list. If the focus is already in the favorites, it jumps directly to the top.

ICONS

Each game supported by MAME can have an associated, freestanding `*.ico` in the `.\icons` directory. IV/Play also supports having all the icons in `\icons\icons.zip` or `\icons\icons.7z`.

Base icon size and the icon scale presets can be chosen with the drop box in **F1**.

If 'Draw non-working overlay' is toggled on in the configuration menu, a small red circle 'x' will be overlaid on any non-working game in the lower right-hand corner of its icon.

ART AREA

The upper right of the game list is the art area. The art area displays the current art type for the selected game. These are freestanding `*.png` files that reside in their respective directories and take the name of each game, e.g. `dkong.png`. IV/Play also support archived art assets so `\snap\snap.zip` or `\snap\snap.7z` are viable options.

Alt-1 through Alt-0 toggles the various art types (which are added through the **F1** configuration dialogue).

Opacity for the art types and art border is set in **F1** configuration dialogue.

The border width and color for the art can be set with **F1**.

Left clicking on the art area with the cursor toggles the art type.

If a game is displayed for which there is no parent snapshot (for example after a new version of MAME is released and linked, prior to a snapshot pack being created), then nothing displays in the art area, the bkground.png will show through.

IV/Play supports History.XML and MAMEInfo.DAT support files available at their respective sites. They are searched for in the MAME.exe directory structure on initial IV/Play launch and can be added at any time through the **F1** configuration menu. These files' information is accessible with the 'Peek' option using the tilde key (~) above the tab on most keyboards. Left and right in DAT peek mode sizes the peek overlay horizontally, the **TAB** key cycles through the DAT types while the peek overlay is open.

Snapshots and other art assets are placed in the upper right of the window offset from the edge, and are scaled to the height of the window. If art is wider than 40% of the horizontal width of the window, it will be scaled to fit. Border width, color, and its opacity are set in the **F1** screen.

TEXT FILTER DIALOGUE

The Text Filter dialogue box is accessible with either **Ctrl-F** or **Ctrl-E**, and is extremely flexible. When typing in this field and hitting enter it will search and display resulting hits, like a filter. For example, typing 'donk' would bring up Donkey Kong and all other items with 'donk' in its description including something like 'Crazy Donkey'.

The filter can also take manufacturer, year, driver source for the game, i.e. 'dkong.cpp', and other key elements. Filtered game list is maintained until the filter is cleared. The filter includes favorites if toggled on. The dialogue can be dismissed with **ESC**. Filtered result counts are displayed on the title bar. A filtered view may be backed out of with **ESC**. The filter can also take powerful advanced searches and gives some examples in its drop down. It is able to form these advanced searches based on elements imported from the MAME XML. See **Appendix 1** for additional filter syntax and information on natural language 'aliases' for some common search topics. **Note:** The filter dialogue opens a tiered Quick Start dropdown automatically when the field is empty — 12 curated filters appear immediately, with the full library accessible via Alt-Down or the hint line. Down arrow or clicking the blank filter area also opens it. The most recent user filter appears above the Quick Start entries. To disable the two-step discovery type filter dialogue, chose the 'Use Classic Filter' in **F1**.

BACKGROUND SELECTION AND RANDOMIZATION

The bottom art layer of the main list area takes a user defined background *.png image from the .\bkground\ directory. The **F1** configuration dialogue toggles the background image rotation, so IV/Play will use a random *.png from the .\bkground directory on each launch. **Note:** If the bkground image is 'larger' than current game window it will scale to fit the inner window. If the bkground image is smaller than current game window and less than 903px wide and 800 tall, it will tile from upper left. These bounds can be altered in the *.cfg.

PROPERTIES VIEW

Alt-ENTER displays the properties dialogue for the selected game. It can be dismissed with the **ESC** key. The text in each field is highlight-able and can be copied.

LAUNCH GAME

Launching a game via Enter or double-clicking sends the simple argument 'MAME.exe dkong' to the linked MAME.exe. The configuration for MAME is done using its own *.ini files. Holding down the **Ctrl** key and hitting enter will launch the game and run its last recorded *.inp file if found. By default, a regular launch will record an *.inp of the game session. Holding down **SHIFT** and hitting enter on a software list machine will launch that machine as if it had no cartridge inserted, like an Atari 2600 or Intellivision.

ACCELERATOR / NAVIGATION KEYS

The following are the keyboard shortcuts in IV/Play:

~ (Tilde)	Toggles the DAT peek feature. Use with left and right arrows to size from 40 to 75 and 100 percent UI width. Up, down, Page Up/Down, Home, and End can be used to scroll, as is common in the other text areas of the app. The TAB key will move to the next DAT file and then loop.	User Guide
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Alt-1 through Alt-0	Go directly to art type. (left click on art area) Note: 'None' is always 0.	
Alt-D	Toggle the display of favorites.ini. (off, favorites + games, favorites only)	
Alt-Enter	Display properties for the game selected.	
Alt-I	Cycle through the icon display sizing presets.	
Alt-Insert / Delete	Cycles through Custom Game Lists (e.g. Full, Arcade, User Lists)	
Alt-L	Save the current visible list as a new Custom Gamelist entry.	
Alt-Left Arrow	Exits a DAT overlay, software list view, or clears and active filter.	
Alt-O	Select a new random bkground.	
Alt-P / TAB	Toggle through the different game list views. (right click on UI)	
Alt-S	Cycles through the main 5 sort options while in flat view.	
Arrow Left / Right	Navigate one letter's worth of games, from A to B to C, etc. through favorites and down through the main game list eventually looping to the top again.	
Arrow Up / Down	Navigate one game at a time.	

Ctrl-B	Benchmark the currently selected machine; default is 90 emulated seconds.
Ctrl-D	Add or remove a favorite from favorites.ini.
Ctrl-Enter	Launch the inp of a game that has been previously recorded. (note that by default any game launched normally will have its session recorded as an .inp)
Ctrl-F	Open text filter dialogue.
Ctrl-R	Select a random game.
Ctrl-Shift-0	Re-center and reset the dimensions to default without touching any other settings.
Ctrl-Shift-C	Copies the contents of the filter results list to the clip board in tab separated, Excel friendly format.
Ctrl-Shift-L	Cycle command line history in F1.
F1	Display the configuration dialogue.
F10	Internal tool for Build Verification Test.
F11	Toggles full screen mode, removes title bar and window frames.
F12	Build the available list based on zipped ROMs found in the MAME.ini rompath.
F2	Launches the IV-Play.log overlay.
F3	Launches the IV-Play.cfg overlay.
F4	Select the MAME.exe.
F5	Refresh the XML import from MAME.exe and rebuild icon / art caches.
F6	Internal tool for art asset auditing.
F7	Toggle the onscreen diagnostic dashboard for FPS, memory usage for CPU and GPU, and garbage collection.
F8	Factory reset, removes the \IV-Play directory and *.cfg but leaves favorites.ini and the IV-Play.Custom.ini. It will prompt for the action. Since it removes the available .bin file a re-audit is necessary with F12 once completed if that feature is being utilised.
F9	Runs the BigBenchmark suite of 36 games for 90 emulated seconds each and reports to console and saves a text file to the user desktop.

Home / End	Move to the beginning / end of the game list, or as a stop at the beginning and end of the favorites (press home/end a second time to navigate out of favorites).
Ins / Del	Cycle through the dynamic themes while in the game list.
Page Up / Page Down	Navigate one window height worth of games.
Shift-1-9-Enter	Launches the selected game in N concurrent tiled windowed instances.
Shift-Alt-S	Cycle sort direction in Flat View.
Shift-Enter	Launch a softlist machine without entering its softlist. (effectively like powering on a console with no cartridge in it)
Shift-F8	Display a grid of all the IV/Play splash screens.

EXTERNAL FILES

IV/Play.exe can reside in its own folder or along with MAME.exe. It also now creates a sub-folder for its asset files for easy rebuilds. It's recommended to keep the *.inis at the same level as the .exe so they survive directory purges.

C:\games\IV-Play\IV-Play.exe (main application)
C:\games\IV-Play\IV-Play User Guide.pdf (this document)
C:\games\IV-Play\IV-Play\IV-Play.sqlite3.dll (required SQLite native library)
C:\games\IV-Play\Favorites.ini (user favorites list)
C:\games\IV-Play\IV-Play.Custom.ini (custom game list definitions)
C:\games\IV-Play\IV-Play.cfg (application configuration)

C:\games\IV-Play\IV-Play\IV-Play.available.bin (current results of F12 available ROMs audit)
C:\games\IV-Play\IV-Play\IV-Play.background.state.json (background rotation tracking)
C:\games\IV-Play\IV-Play\IV-Play.db (SQLite game database)
C:\games\IV-Play\IV-Play\IV-Play.db.meta.json (database schema metadata)
C:\games\IV-Play\IV-Play\IV-Play.history.cache (parsed History.XML binary cache)
C:\games\IV-Play\IV-Play\IV-Play.icon.atlas.png (combined icon texture atlas)
C:\games\IV-Play\IV-Play\IV-Play.icon.map.json (icon atlas coordinate index)
C:\games\IV-Play\IV-Play\IV-Play.icon_atlas.hash (atlas validity checksum)
C:\games\IV-Play\IV-Play\IV-Play.log.txt (diagnostic log)
C:\games\IV-Play\IV-Play\IV-Play.mameinfo.cache (parsed MAMEInfo.DAT binary cache)
C:\games\IV-Play\IV-Play\IV-Play.SoftListFavorites.cache (softlist favorites binary cache)
C:\games\IV-Play\IV-Play\IV-Play.SoftlistIndex.cache (140K-item softlist binary cache)

Once linked to a MAME.exe, IV/Play will use its structure for its art, for example:

C:\games\MAME\MAME.exe
C:\games\MAME\History.XML
C:\games\MAME\MAMEInfo.DAT

C:\games\MAME\INI\MAME.ini
C:\games\MAME\bkgground\bkgground.png
C:\games\MAME\icons\gamename.ico
C:\games\MAME\snap\gamename.png
C:\games\MAME\flyers\gamename.png
C:\games\MAME\etc\etc.png

IV-PLAY.CFG

This file is created on first launch of IV/Play if it is not present. It contains the configuration settings for the application. Delete this file (and the IV-Play sub-directory) to start fresh. Note that the *.cfg file can point to a custom asset directory (asset_directory), which is helpful if a RAM drive or similar technology is being employed for storage.

IV-PLAY.DB

The DB file is an SQLite database created from the contents of the XML output produced by the MAME engine. It is created on first run if not present after asking for the .exe. If IV/Play is run without the *.db in its directory it will ask for a location of the MAME.exe (if not finding it in the same directory) and use that path in the IV-Play.cfg file. **F5** refreshes the XML and updates the *.dat. **F4** re-links to a new MAME.exe. Delete this file to affect a re-build of the database. **Note:** the **e_sqlite3.dll** **must** be in the same directory as the IV/Play .exe for the app to work.

UPDATES

IV/PLAY 1.5 FEATURES

UNLIMITED ART TYPES

IV/Play allows for the display of 'unlimited' art types. In the **F1** configuration dialogue, any directory can be added that contains art types following the *gamename.png* format. IV/Play will automatically add snap, flyers, cabinets, PCB, marquees, cpanel, titles, bosses, ends, gameover, logo, scores, select, versus, howto, warning, and artpreview on initial launch if it finds them (including their respective *.zip or *.7z files). The view order can be set by moving the folders up or down the list. The art types will be assigned **Alt-<X>** shortcut keys depending on position, **Alt-1** through **Alt-0**.

HISTORY.XML / MAMEINFO.DAT SUPPORT

IV/Play supports History.XML and MAMEInfo.DAT. They are searched for in the MAME.exe directory on initial IV/Play launch and can be added at any time through the **F1** configuration menu. Activation of the view is via the tilde ~ key which opens the DAT 'peek' view. The peek overlay is expanded left or right to multiple sizes and navigation is via the regular arrow keys, **page up** and **down**, and **Ctrl-End** and **Ctrl-Home**. A separate font with size and color is selectable for the text area on the **F1** configuration menu.

FAVORITES TOGGLE

Favorites can be cycled with **Alt-D** through off, on with game-list, and favorites only. This also works in softlist view and in filter results.

NON-WORKING GAME ART/HISTORY SUPPORT

IV/Play will display icons and any art type for non-working games if they are present. This also allows viewing of History.XML and MAMEInfo.DAT entries for non-working games.

COMMAND LINE OVERRIDE

The command line override in the **F1** configuration dialogue allows the use of various switches to be added to launched games, e.g. -window to play games windowed without having to drop to the mame.ini for editing. Note that command line arguments take precedence over MAME's *.ini processing hierarchy. The command line is now a drop down and has a well with the last 5 commands remembered. The command lines can also be cycled through with **Ctrl-Shift-L** and will complain if the command does not begin with MAME's expected dash (-) prepending the arguments.

IV/PLAY 1.6 FEATURES

Updated to work with new XML output of MAME .162.

IV/PLAY 1.7 FEATURES

CLONE AND NON-WORKING FILTERS

IV/Play now has the ability to filter out clones and/or non-working machines in the **F1** configuration screen.

IV/PLAY 1.8 FEATURES

- New background and modern Windows Segoe UI font as default on first run.
- ESC will exit IV/Play.
- It is now possible to un-indent the clones in **F1**.
- Relative paths are now used in the IV-Play.cfg file, this allows for easy portability/transport via thumb drive or removable storage.
- With 1.8.4.0 added support for the new format of the History file, now an *.XML at www.arcade-history.com.
- With 1.8.5.0 added the ability to change the color of the non-working games/devices in **F1**.

IV/PLAY 2.0 FEATURES

After 9 years in maintenance mode, IV/Play sees a significant modernization effort, rebuild, major feature additions, and a full version bump in the summer/fall of 2025.

MOVE FROM .NET 4.5 TO .NET 8.0 & EXIT FROM WPFWRAPPER TO PURE WINFORM

A migration was undertaken successfully to get IV/Play from .NET 4.5 all the way to 8.0 and also to extricate it from the WPFWrapper which had ultimately caused bothersome flickering of the handoff on launch.

GPU RENDER PIPELINE

IV/Play has been converted from a GDI app to utilising a full GPU accelerated pipeline. This leads to considerably smoother scrolling and navigation in-app with art assets appearing instantly from its various caches. Art, icons and backgrounds scale via DirectX High Quality Bicubic instead of straight bilinear filtering, resulting in sharper images on higher resolution monitors.

This architectural change was a strategic decision to resolve a persistent and severe performance bug known as "scrolling judder". The previous GDI+ rendering pipeline was CPU-bound, meaning all drawing operations were handled by the main processor. During fast navigation, this would overwhelm the UI thread, causing the application to stutter or freeze.

The new pipeline offloads all rendering to the system's graphics card (GPU) using DirectX 11 / Direct2D. This frees the CPU from heavy drawing tasks, resulting in a perfectly smooth, zero-latency user experience, even when scrolling rapidly through thousands of games with high-resolution artwork. This one-time processing of assets on startup is an intentional trade-off that enables the application's highly responsive and fluid feel during use.

LITEDB TO SQLITE

To ensure long-term stability and performance, IV/Play's data layer was migrated from the LiteDB engine to SQLite using the powerful Entity Framework Core.

This was a critical architectural change driven by a series of persistent and difficult-to-diagnose crashes originating from the LiteDB query engine. The migration to a mature, industry-standard backend like SQLite completely resolved these stability issues.

SUPPORT FOR HIGH DPI MONITORS / LARGE ICON MODE / SCALING IMPROVEMENTS

- IV/Play is now high DPI aware; it won't be necessary to use Windows 11 compatibility hacks to enable a non-miniscule screen. The UI is now at 1523x648 which is proportionally what it was at 1015x432 on 1080p monitors. Running on 1440DPI and 4K now looks very nice; the fonts are crisp and each art asset looks the way it should instead of being simply stretched after a Windows snapshot. Everything scales properly based on the DPI setting including row height.
- Large icon view is now supported, it can be toggled in the **F1** dialogue and via **Alt-P** or **TAB** as one of the views in the cycle.
- Backgrounds will now scale themselves to the inner window size, a set size floor for that functionality can be altered in the *.cfg so anything smaller than the setting will be tiled. This works particularly well for aspect ratio friendly bkgounds that may be slightly bigger or smaller in their dimensions than the current inner window.
- Long time bug fixed for maintaining a maximized state on re-launch.
- IV/Play no longer locks its assets with file handles, so the snaps viewed in that session can be replaced on disk underneath the app.

SUPPORT FOR SOFTWARE LIST DEVICES

IV/Play can now click/enter on a console like Atari 2600 and it will read its softlist and display as the new game list. The navigation is the same as the full game list. Hit **Backspace** or **ESC** or **Alt-Left-Arrow** to return to the level above with highlight on the system entered. The favorites work the same way in the softlist view, adding or removing them with **Ctrl-D**, or toggling their views with **Alt-D**. Use **Shift-Enter** to launch a softlist machine directly with no software/cart attached, effectively like powering on an Atari 2600 with no cartridge in the slot.

CUSTOM GAME LIST SUPPORT

Similar to the favorites.ini, IV/Play now allows the use of the IV-Play.Custom.ini file, selectable from the **F1** dialogue. If chosen, this file takes over the game list and displays entries contained within. This is a single file, but it can be nested based on **[]** or **<>** heading delimiters that will produce the drop-down choice.

```
#Custom GameLists
```

```
<Golden Age>
```

```
amidar
```

```
armora
```

```
astdelux
```

```
asteroid
```

```
<Fighters>
```

```
sf2
```

```
sfa3
```

GRID VIEW

Added the grid view. This will display the games in a grid with machine names under the icons for readability. The grid view behaves in the same fashion as the Game list view. **Alt-P** and **TAB** will cycle through the game list modes: regular icon list, large icon list, large icon grid, large icon grid + no art view.

BACKGROUND BRIGHTNESS DETECTION MODE

IV-Play can now automatically determine if the bkground image is of a very light/bright variety and adjust its fonts to black accordingly (which can still be overridden in the **F1** config).

IMAGE ATLAS FOR ICONS AND BINARY VERSIONS OF HISTORY.XML AND MAMEINFO.DAT

To maximize performance, IV/Play combines all individual game icons into a single large image file called an "icon atlas". This atlas is loaded into GPU memory on startup, allowing for extremely fast icon display and smooth scrolling by eliminating the need to load thousands of individual files from the disk.

To accelerate startup, IV/Play creates optimized cache files for History.XML and MAMEInfo.DAT the first time they are used. On all subsequent launches, the application reads from these pre-parsed files, dramatically reducing load times by avoiding the slow process of reading the large, original text files.

ADDITIONAL FEATURES

- IV/Play.cfg is now alphabetized and sectionalized.
- Via the auto-record **F1** toggle, machines launched will have their session *.inp recorded. To play it back, launch the same game a second time with the **Ctrl** key held down.
- Nearest neighbor scaling is an option now in **F1** (unsmoothed snapshots); it allows a more pixelated scaling algorithm that lends a retro 'air' to the snaps.
- A simulated scanline effect can now be applied to snaps, both over the pixelated and the smooth versions.
- Clone icons and the focus/highlight are now transparent and controlled via settings in the *.cfg file.
- Launching fresh on a machine with a 1080p monitor will alter the DPI settings to the historical dimensions of 1015x432 and alter the icon and row size appropriately. Delete the *.cfg file or Reset to Defaults in **F1** if IV-Play has travelled portably from a 4K system to reset to the 96dpi destination.

IV/PLAY 2.1 FEATURES

MOVE FROM .NET 8 TO .NET 9

Upgraded the project from .NET 8 to .NET 9 to benefit from the latest runtime performance enhancements, contributing to a more responsive UI and faster icon cache processing.

ADDITIONAL FEATURES

- Tapping the **Tilde (~)** key will open the DAT 'peek' view where the DAT files are accessible in a temp window. Left and right size the window to 40%, 75%, and full width of the UI. **TAB** cycles through the DATs.
- Copy out of Filter, create a filter search and then **Ctrl-Shift-C** to copy the results to the clipboard in a tabbed, Excel friendly output.
- **F2** (log) and **F3** (config) now activate their 'peek' overlays with navigation the same as the DAT peek.
- **F8** will do a 'factory reset' and remove the contents of the \IV-Play directory including the *.db and art caches. It also removes the *.cfg file, but leaves the favorites.ini and custom game list ini.
- Aspect ratio constrained sizing: If **Ctrl** is held down during a sizing event the UI will scale up or down in 2.35:1 TohoScope.
- **F7** displays the performance dashboard/overlay in the upper right corner giving information on memory usage for the system, GPU, FPS, and garbage collection data.
- Distributable is now packaged with AOT (Ahead-of-Time) compilation during publication (vs. JIT) which speeds the initial bring-up of the app.
- **F10** runs a short BVT (Build Verification Test) used in development to exercise basic functionality for breakage.
- Added a 'Theme' drop down to **F1** which allows a single switch for multiple color combos on the various game list entries.
- Added ability to launch a softlist machine on its own, e.g. 'mame.exe a2600', instead of going into its softlist view. Hold the **Shift** key down while hitting enter or double clicking on the machine. This is akin to starting a console with no cart in it.

- Reintroduced an **F1** toggle to hide non-working mechanical machines, removing ~15K entries from the game list; this is now on by default.
- Added a drop down in **F1** for icon scale presets based on monitor resolutions and DPI combos. This controls the scaled size of the regular and large icon display on said monitors.
- **Alt-I** in the game list will cycle through the various icon scale presets.
- **F11** will now toggle into and out of full screen mode, hiding the window borders and title bar; similar to **F11** in browsers. **ESC** will also exit full screen mode. A default setting for this is user configurable in the *.cfg file in the window section.

IV/PLAY 2.2 FEATURES

VERTICAL SCROLLBAR FOR GAME LIST

A custom vertical scrollbar control has been added to the right of the art area.

FILTER IMPROVEMENTS AND ADDITIONS

The **Ctrl-F/E** filter has been overhauled with a clearer presentation and expanded functionality. Key features are now surfaced in the drop-down example list. Opening the filter on an empty field now shows a **Quick Start Filters** preview tier: 12 high-value curated filters shown immediately, with the full library accessible via Alt-Down or by clicking the hint line at the bottom of the list. Natural language terms such as *80s games*, *working only*, or *no clones* can be entered directly, while the full SQL-like advanced syntax remains available for precise control with field-specific searches (e.g., year:1994), negation (manufacturer:!Capcom), and complex Boolean logic. New hybrid queries now allow natural language keywords to be combined with advanced operators in a single expression, enabling flexible searches such as Year >= "1995" noclones for parent games from 1995 or later, manufacturer:Capcom working for working Capcom titles, or description LIKE "%puzzle%" nonworking parents for non-working parent puzzle games. This hybrid approach bridges simplicity and precision, giving both casual and advanced users greater control over filtering results.

ADDITIONAL FEATURES

- IV/Play's version information is now surfaced on the title bar after its name before the linked MAME .exe.
- The DAT Peek feature's key is now configurable; especially handy for non-US keyboard layouts. The default will try to find the key in the upper left above the **TAB** key and should handle most European keyboards. This key can be set in the IV-Play.cfg file. The various key codes may be found on Microsoft's site by searching for System.Windows.Forms.Keys enumeration in their .NET Framework docs.
- The **F1** config check boxes have been re-ordered and the verb has standardized on 'Show' vs. a mix of show and hide. Attendant logic has been updated for the new verbiage.
- The Grid View + No Art combo has been added to the **TAB / Alt-P** cycle as the fourth option.
- The diagnostic **F10** Build Verification Test (BVT) has been extended with a new overlay and progress screen with grid view interactions.

IV/PLAY 2.3 FEATURES

REINTRODUCED LEGACY ERROR DETECTION

Reintroduced the legacy IV/Play (1.8.5) feature that detects and reports MAME errors when attempting to run a machine without the required ROMs. This is displayed via one of the overlays and allows a copy and paste out of.

BIGBENCH BENCHMARK SCRIPT

Added John IV's *BigBench* benchmark script. Launchable via **F9** (and dedicated **F1** button), it runs 36 games in succession, measuring performance over 90 (emulated) second intervals. Monitoring overhead is negligible, and results are statistically equivalent to those obtained from a direct command-line run. Results are displayed in an overlay and also saved as a text file on the user's desktop. Both the game list and the -bench (time) value are configurable in the *.cfg file. See the benchmark page at the IV/Play site for historical data: <https://john-iv.github.io/iv-play/bench.html>. Note that benchmark results for each new MAME release are typically posted monthly on the [MAMEWorld News](#) forum. The geometric mean value is the current preferred measure of PC / MAME performance across disparate hardware.

DYNAMIC COLOR THEMES

Added dynamic theme options in addition to the static presets accessible in the **F1** config. Dynamic themes calculate text and UI colors based on the dominant background image color and its brightness. This allows the interface to adapt automatically to different backgrounds while maintaining legibility and acts on the fonts, scrollbar thumb, and art border. **Insert** and **Delete** cycle through the dynamic themes while in the game list.

- **Brightness Ladder** Adjusts text lightness relative to the background. On darker backgrounds, text colors are brightened in stepped increments to ensure readability. On lighter backgrounds, text colors are darkened to maintain contrast.
- **Saturation Gradient** Maintains a fixed lightness level but varies saturation. Favorites are rendered with higher saturation, clones with reduced saturation, and non-working entries in grayscale. This produces a consistent brightness level while distinguishing categories by color intensity.
- **Analogous Palette** Selects colors close to the background's hue, shifted slightly on the color wheel. A minimum lightness threshold is enforced to prevent text from becoming too dark. This produces a cohesive palette aligned with the background color.
- **Dual Contrast** Applies both saturation and brightness adjustments to create stronger separation between categories. Favorites and parents are shifted lighter and more saturated, while clones are darker and less saturated. Non-working entries are rendered in grayscale with a minimum brightness floor to ensure visibility.
- **Miami Pastels** Selects a vibrant, high-saturation color scheme using complementary pastel hues, typically teal and pink. Colors are assigned based on the background's general warmth (warm background triggers cool pastel text, and vice versa). A fixed lightness level is targeted to maintain the signature pastel aesthetic, with slight adjustments made for very bright backgrounds. This theme overrides the hue of the background completely to apply a distinct, stylized color set.

EXPANDED KEY CUSTOMIZATION

Users can now modify several keyboard shortcuts to better suit their preferences. These customizations are made directly within the IV-Play.cfg configuration file. Look for the Key Bindings section near the top of the file; it includes comments explaining the syntax and providing examples of valid key combinations. Formalized the Euro keyboard version of the 'Tilde Peek' key, which is Oem8 and made it a standard alias.

FAVORITES.INI AUDIT

Toggled on by default in the *.cfg, this feature will audit the favorites.ini file on initial launch looking at each line. If it finds something it doesn't recognize it will prepend an asterisk to the entry; to surface it for manual editing and removal. It also displays an overlay with this information in-app. This feature helps keep track of occasional renames of machines that occur in baseline MAME .exe, and also clean up accidental misspelled manual entries to the favorites.ini.

ADDITIONAL FEATURES

- The scrollbar thumb color can now be customized in the F1 configuration screen. The active scrollbar thumb color may also be set manually in the *.cfg file. If left at the default, IV/Play will automatically adjust the thumb color based on background brightness, selecting either a dark or light variant for optimal visibility. The scrollbar thumb also takes the dynamic themes colors if not user customized.
- The border opacity has been surfaced into a control on the **F1** config dialogue.
- Added a toggle for Random Dynamic Theme in **F1**. If checked, this will randomly cycle through the dynamic themes on launch, providing an attractive new look each session; especially when paired with randomizing the background.
- MAME and IV/Play versions now appear in the **F1** config form as well as a link to the home page.
- Benchmark button added to the **F1** config.
- The default short-cut keys will now display in the **F1** dialogue for easy reference.
- Added the ability to tweak which themes are available to the theme cycle and which are included in the randomization carousel in the *.cfg file.
- Added a button in **F1** to set the background images folder.
- **F6** launches an internal tool for icon and snap asset auditing.

IV/PLAY 2.4 FEATURES

SOFTWARE LIST FILTERING & FAVORITES

A large refactoring of the handling of software lists now allows for filtering/searching across the entire 100K plus collection of MAME software items. This means it's now possible to search for an item like name:Zaxxon and get the hits for Zaxxon across all of its ports. Software list items are also now eligible for inclusion to the favorites.ini file. Add or remove them with the same key **Ctrl-D**. They can be grouped at the end of the main game list favorites (sorted by machine) or integrated into the main list via descriptive name. The favorites audit will now extend to include these items, so a2600/combak vs. a2600/combat will be flagged.

F1 CUSTOMIZABLE KEYS SECTION

The main alterable key combos have been surfaced into the **F1** dialogue. All the remaining keys are editable in the IV-Play.cfg file.

THE ARCADE GAMELIST

IV/Play now has a curated, subjective, arcade-only game list; accessible via **Alt-INS** / **Alt-DEL** cycle keys or **F1** in the custom game list drop down. Excludes all the non-coin-op, mahjong, fruit, gambling, bar top, quiz, pinball, redemption, mechanical, and BIOS machines. With clones and non-working filtered out it's about 3400 games.

SUPPORT FOR ZIP AND .7Z ART PACKS

Added the ability to use \snap\Snap.zip|.7z and \icons\Icons.zip|.7z and the other art types, cabinets, flyers, marquees, etc. **Note:** 7z solid archives are not supported due to performance degradation, the app will warn if it finds one in its asset search. The Snap.7z and Icons.7z from the [IV/Play Homepage](#) can now simply be dropped in to the \icons and \snap directories for direct use.

MULTIPLE MAME LAUNCHES AND TILING

IV/Play now allows for using a single session of itself to launch multiple windowed instances of MAME; selecting either different games or the same game multiple times for multi-player gaming on a single screen. Use **Shift-1** through **Shift-9** while launching a game with **ENTER** or double-click to create that many windows. This allows for playing networking supported games in MAME like linked Cruisn' USA cabinets or similar. **Note:** Only the first instance gets the sound, to prevent the cacophony of all of them outputting at once. The first windowed game also takes the mouse vs. the others for easier navigation and windowed placement, as desired.

MOVE FROM .NET 9 TO .NET 10

Upgraded the project from .NET 9 to .NET 10 to take advantage of long-term support and further runtime optimizations, delivering faster startup responsiveness, smoother UI interactions, and improved memory efficiency for a more stable user experience.

ADDITIONAL FEATURES

- The art border size and the opacity settings now have a preview box next to them to visually balance out some of the other additions to **F1**.
- Verbose logging is now a toggleable option in **F1**.
- IV/Play's start-up time is now displayed on the **F1**, this includes time to an interactive UI and the final completion of background threads.
- Added a 5-layer history drop-down to the MAME command-line override combo box in **F1**, with the ability to assign a shortcut key to cycle, currently **Ctrl-Shift-L**, tweak in the *.cfg key mappings section.
- Added a 'User Guide' button to the **F1**, this will launch this document if it is in the same directory as the .exe.
- An individual machine can now be benchmarked; use **Ctrl-B** on a selected item to get the same result the full bench suite produces. The duration is the default 90 emulated seconds. This can be adjusted in the *.cfg file as desired.

- Allow a nonworking machine to display its own art asset if there is one.
- Prevent snap scanlines on vector games and non-MAME generated art (flyers, marquees, PCBs, etc.)
- Allow saving off the random bkground tracking json file before a full factory reset.
- Allow DAT peek with art type 'None'.
- Auto-populate paths for the current common art types if present in the MAME.exe directory structure.
- Added Record INPs and Launch in Fullscreen as toggles in **F1**.
- The app now hides itself when launching MAME.
- Reworked the status line on the game Properties view (Alt-Enter) to provide more detailed and clearer information presentation.
- Altered the link to MAME.exe dialogue to default to all files first (*.*) so WINE users on other hardware are able to more easily attach to a MAME without renaming.
- Added a toggle in **F1** to show or hide Softlist machines themselves.
- Added ability to search by softlist media types like cartridge, cassette, CD-ROM, etc.
- Added autocomplete with 'ghost text' look-ahead to the **Ctrl-F** filter dialogue.

IV/PLAY 2.6 FEATURES

FLAT VIEW & GLOBAL SORT

Added a new flat view which displays all machines alphabetically and non-indented; clones decoupled from their parents, favorites not pinned, parents at full opacity. Accessible via the **TAB** cycle or the **F1** View dropdown. Flat View is also the only mode that supports global sort, which orders the list by default, description, machine name, manufacturer, or year. The selected sort field persists through restarts; direction resets to its natural default. Sort cycles via **Alt-S** or the **F1** Sort dropdown; especially useful when a filter result contains thousands of entries.

DEEPER DAT INTEGRATION FOR FILTER SEARCHES

History.xml, MAMEINFO.dat, and CatList.ini are now indexed/hydrated at first use and integrated into the **Ctrl-F** filter system. history: (or hist:) searches the descriptive text from History.xml, while genre: (gg:) queries CatList.ini categories and subgenres. MAMEINFO.dat contributes first-appearance version data, exposed through mameversion: and mv: along with natural language forms such as "version 53." These sources also participate in the natural-language layer, allowing combined queries like "imperfect shooters from 1995" or "platformers added in 0.99." Two new operators workingstatus: and supportstatus: extend filtering to MAME's internal status flags for both arcade and softlist items. See the down arrow on the Filter dialogue for a series of examples.

AVAILABLE ROMS GAMELIST

The 'Available' gamelist, accessible from the **F1** Custom Gamelist dropdown or via the shortcut cycle key **Alt-INS/DEL**, performs a simple .zip/.7z audit on the rompath entries defined in MAME.ini and matches those against machines in the XML, providing a list of only found items. Detection is non-recursive and name-only. Clones' presence is assumed when using fully merged ROM sets. The gamelist intentionally does not show ROM-less games like Pong; switch back to the full gamelist to see those entries. To activate this feature it is necessary to hit **F12** to do a quick audit of archival ROMs in MAME's -rompath.

PLATFORM-BASED SOFTLIST FILTER GROUPING

When a filter (**Ctrl-F**) returns software list results, IV/Play now collapses them into platform-level family nodes rather than producing one row per software item per hardware variant. A search for 'zaxxon' produces a single 'Zaxxon (Atari 2600)' entry, a single 'Zaxxon (ColecoVision)' entry, a single 'Zaxxon (MSX1)' entry, and so on; one row per platform, regardless of how many hardware machine variants or regional releases exist for that platform in MAME. Pressing Enter on a family node drills down into it, showing the parent release and any clones (regional variants, alternate revisions) for that platform. The parent entry shows its clone count, and each clone shows its parent name in brackets. Press **Alt-Left Arrow**, **Backspace**, or **ESC** to return to the platform-level filter results. Adding 'noclones' to the filter suppresses clone entries before grouping runs, so only parent releases appear and drill-down shows a single launchable entry.

NATIVE AOT TRANSITION / PERFORMANCE

IV/Play now ships as a native Ahead-of-Time (AOT) build, producing a smaller, fully self-contained executable that no longer requires users to install the .NET 10 runtime (**Note:** the .exe size increases as a byproduct). Startup behavior is more consistent across machines, with faster warm launches and smoother recovery after full XML exports or database rebuilds. Overall responsiveness improves due to native code generation, and memory usage is slightly reduced.

MACHINE STATUS AWARE ICON BORDERS

This configuration feature surfaces the status of MAME machines that appear in **Alt-Enter** properties as not working (red), imperfect (orange), and working (green) by using the black border of the existing icons and changing them to that color (or any user chosen in the *.cfg). Further, since the imperfect games are the most in need of visual differentiation, they are now defaulted to a background derived themed color to complement the auto-theme feature. This auto color for the imperfect status border can be turned off in **F1** to return to user chosen RGB settings. It also extends to softlist machine media items.

HIDDEN.INI

IV/Play now supports a hidden.ini file at the executable level for permanently suppressing entries from the game list before any filter or display logic runs. Each line takes one of three prefix forms: type: targets a machine category, machine: targets a specific short name, and driver: targets all machines sharing a source driver. Excluded entries are removed from the dataset entirely — they do not appear in filter results, favorites, or any custom gamelist. The file follows the same precedence rules as Favorites.ini and survives factory resets. It is generated automatically if not present and contains the syntax/usage information. Its creation can be suppressed by a toggle in the *.cfg if desired.

CUSTOM GAME LIST GENERATION FROM FILTER RESULTS

Using **Alt-L** from a filter result set will now automatically create an entry in the custom-list.ini file.

NUMBER OF CLONES / PARENT

Added the ability to show the number of clones a machine has on the gamelist, as well as the counterpart '[parent: x]' on clone entries. Clone counts use singular/plural forms ("1 clone" / "5 clones") and appear left-anchored

immediately after the machine description. This can reveal interesting historical patterns where widely bootlegged games may have dozens of clones. Accessible in **F1** as two independent toggles.

COMMAND LINE OVERRIDE AUTOCOMPLETE

Returning from IV/Play 1.8, MAME's ``-showusage`` output is captured and used for the command line override text input.

ADDITIONAL FEATURES

- DPI detection and display has been hardened; IV/Play now flows better when dragged across multiple monitors with different DPIs and scaling settings.
- Added toggles for 'Show Parents' and 'Show Machines' to create even more display permutations. These can also be accessed via natural language queries in **Ctrl-F** such as "Parents" or "Machines Only."
- Added new randomized splash screens which appear on full MAME XML export and .db rebuilds at launch.
- Added **Shift-F8** to show all the new internal splash screens in an overlay panel.
- Added two new 3-way filter operators to **Ctrl-F**: `workingstatus:working / :imperfect / :notworking` (alias `ws`;) distinguishes fully-working from imperfect-emulation machines, going beyond the boolean `working only` phrase. `supportstatus:yes / :partial / :no` (alias `ss`;) filters softlist items by their MAME support level. Combine with a system name for best results: `snes supportstatus:yes`. Natural language aliases such as `imperfect machines`, `broken games`, and `partially supported` are also supported.
- Added a 'snapshot' of the entire user configuration taken on launch and available in the **F2** log. If changes are made through the session, they are flagged by a color change and delta icon (Δ) on the next **F2** to make for easier debugging/diagnostic work.
- Added Genre, MAME version, and machine type to **Alt-Enter** properties since we consume all of that and hydrate it now. Requires History.xml, MAMEINFO.dat, and CatList.ini.
- Added a subtle configuration/**F1** button at the bottom of the vertical scrollbar. This is the first time in the app's history that the configuration dialogue is accessible without a keypress.
- Added current game list in use to the titlebar.
- Add **Alt-O** to select a new random bkground. **Note** shortcut keys are all assignable in the *.cfg and some major ones in the **F1** dialogue.
- Rename favorites and custom gamelist in the distribution packages so they don't overwrite users' own files.
- Added the ability to return the app to center of screen and its default dimensions. Double click the titlebar or use **Ctrl-Shift-0**.
- Added an OS aware System / Dark / Off mode drop-down in **F1**.
- Added a browser-like **Ctrl-F** (Find) inside the overlays, including **F2** & **F3**, and the DAT peeks.
- Modernized the font and color pickers in **F1**.
- Removed hardcoded Segoe UI usage throughout the rest of the app, main UI font now set in the **F1** config and will propagate. New default per Windows 11 is Segoe UI Variable Text.
- Created a TTF font based on MAME's very old uismall.bdf for use in MAME's internal UI. Double-click it, choose 'Install' to bring it into Windows 11, then use `'-uifont MAMEUISmallMod'` in the MAME.ini to utilise it, in-game. It also looks nice as the UI font in IV/Play itself, set in **F1**. Added to the distribution package.

- Added a third font picker field in **F1**, this for the rest of the UI, including dialogues. This allows a separate gamelist font setting to be decoupled and individualized.
- No longer including favorites.ini and the custom gamelist.ini in the distribution package so there is no accidental overwriting on decompressing the archive.

IV/PLAY 2.8 FEATURES

DPI & LAYOUT MODERNIATION

IV/Play's DPI system has been rebuilt to eliminate the mix of pixel-based and DIP-based sizing that earlier versions relied on. All layout calculations now flow through a unified DPI-aware geometry model, so the interface scales cleanly on high-resolution displays and behaves correctly when moved between monitors with different scale factors. Window metrics, icon sizes, and hit-testing now update deterministically when DPI changes, preventing the fractional drift and inconsistent sizing that could occur before. The result is a sharper, more stable presentation on modern 4K and mixed-DPI setups without requiring any configuration changes from the user.

INTERNAL HISCORE.INI

Added support for an app-specific 'HiScore.ini' that sits next to the .exe. This is one game per line, tab delimited: machine name, tab, hiscore, tab, date. When present it will be displayed in the art area border under the snapshot at 50% opacity; it's subtle. The hiscore in the row can be a combo of numbers and letters so it can take something like '150,000 Level 6'. It can be placed left, right, or centered via the *.cfg setting.

MAME SOURCE .CPP ADDED TO DAT PEEK

Added a link to the MAME source to the DAT Peek feature. The DAT peek cycle will include the machine in question's XX.cpp file so it is easily accessible for informational purposes. MAME source path is set in the *.cfg source_root line.

ADDITIONAL FEATURES

- Added TohoScope Internal / External to the *.cfg. This is an amusing little feature that will alter the dimensions of the main UI to conform with TohoScope's 2.35:1 aspect ratio in two different ways. The current default method and the one that has been used since IV/Play's inception is 2.35:1 for the *entire* app including the Windows chrome. That always meant the internal window area was not TohoScope as a result. The new setting 'Internal' will now make the inner window area 2.35:1, increasing the total height of the app by the titlebar size.
- Live ROM availability monitoring via FileWatcherService. When a ROM is added or removed from a configured ROM directory, the Available gamelist updates automatically approximately one second after the file settles, without requiring an F12 manual scan.
- Auto-refresh of favorites sync on external favorites.ini changes, also via the FileWatcher pipeline.

APPENDIX 1 - FILTER & SEARCH GUIDE

The filter bar (Ctrl-F or Ctrl-E) accepts plain English phrases, keyword modifiers, and a full SQL-like advanced syntax. All three modes can be freely mixed in a single query.

SIMPLE TEXT SEARCH

Type any word or phrase. Multiple words are combined with AND automatically.

pacman midway	Games containing both 'pacman' AND 'midway' anywhere in their data
"metal slug"	Exact phrase match — enclose in double quotes
capcom fighting	All Capcom fighting games (both words must appear)
snes zelda	Softlist system + keyword — finds Zelda titles for SNES

SYSTEM NAME ROUTING

When a recognized system name (such as nes, snes, or c64) appears in a query, IV/Play applies a three-tier routing model. The word that follows the system name — the qualifier — determines which tier of results is returned.

nes	Both the NES hardware driver AND all NES softlist media (default — no qualifier)
nes machine / nes hardware / nes system	Only the NES hardware driver entry (Tier 1). Qualifiers: machine, hardware, system, platform, driver, host
nes games / nes titles / nes software	Only NES softlist media — no hardware driver (Tier 3). Qualifiers: games, titles, media, software, items, library, catalog, collection
nes cartridges / nes floppies / nes cdroms	NES softlist media filtered by media type. Any media-type word (cartridges, floppies, cdroms, cassettes, hdds, memcards) triggers this rule
softlist systems / softlist devices	All hardware drivers that expose a software list, across all platforms (Tier 2)
software only / media only / softlist	All softlist media across every platform, no hardware drivers (Tier 3)

machines only / no softlist	All hardware drivers only, no softlist media (Tier 1)
arcade media	ROM-based arcade machines (MediaType = rom). Distinct from arcade only, which uses the IsArcade flag

Note: a bare system name with no qualifier shows both tiers. Add a qualifier word to pin the result to one tier. These rules apply even when combined with other filters: nes working cartridges restricts to working NES cartridge softlist items.

KEYWORD MODIFIERS

Add these keywords to any search — or use them alone. Keywords can be combined freely.

WORKING / CLONE / PARENT STATE

working only	Show only working games
nonworking / nonworking only	Show only non-working games
imperfect	Show games with 'imperfect' driver status
parents only	Hide all clone games
no clones / noclones	Hide all clone games (aliases)
clones only	Show only clone games (overrides global hide setting)
not parents / no parents / nonparents	Show only clone entries (inverse of parents only)
working parents	Working non-clone games only
nonworking parents	Non-working non-clone games only
working clones	Working clone games only

nonworking clones	Non-working clone games only
working clones of nonworking parents	Playable clones whose parent set is broken
nonworking parents with working clones	Broken parents that have at least one working clone

MACHINE WORKING STATUS (3-WAY)

Use `workingstatus:` (alias: `ws:`) to distinguish the three MAME driver statuses precisely. Note: `working` only and `nonworking` use a boolean (`IsWorking`), which treats good and imperfect both as “working”. Use `workingstatus:` when you need to isolate imperfect from fully good machines. Not available for `softlist` items.

<code>workingstatus:working / ws:working</code>	Fully working only — <code>DriverStatus</code> = “good”. Also: fully working games, fully working machines
<code>workingstatus:imperfect / ws:imperfect</code>	Imperfect emulation only — <code>DriverStatus</code> = “imperfect”. Also: imperfect machines, imperfect games
<code>workingstatus:notworking / ws:notworking</code>	Not working / preliminary — <code>DriverStatus</code> = “preliminary”. Also: broken machines, broken games, not working machines, preliminary

CONTENT TYPE

<code>softlist / software / software only</code>	Show only software list items (console/computer games)
<code>machines only / no softlist</code>	Show only arcade/computer machines, no software items
<code>softlist devices / softlist systems</code>	Machines that have a software list attached
<code>arcade only / coin-op / coinop / coin op / arcade games</code>	Coin-op, non-mechanical, non-redemption video games only
<code>non-arcade</code>	Everything except coin-op arcade machines

bios only	Show only BIOS entries
no bios	Hide all BIOS entries
mechanical	Show mechanical games (pachinko, pinball, etc.)
working mechanical	Working mechanical games only
not mechanical	Exclude mechanical games
devices / device only	Show only MAME device entries (sub-boards, BIOS helpers — not launchable)
no devices / exclude devices	Hide all MAME device entries from results
console / home consoles	Home console hardware (machinetype:console)
computer / home computers	Home computer hardware (machinetype:computer)

FAVORITES

favorites only	Show only games in the favorites list
exclude favorites	Hide all favorites from results
not favorites	Alias for exclude favorites
nofavoriteslist / no favoriteslist / noflist	Suppress the pinned Favorites group at the top of the list without removing favorites from results — creates a single flat list. Useful with broad filters: softlist noflist, capcom noflist

HARDWARE & DISPLAY

vector	Vector graphics games (Asteroids, Tempest, etc.)
raster	Raster (pixel-based) graphics games
vertical / portrait	Vertically oriented monitor games
horizontal / landscape	Horizontally oriented monitor games
chd games	Arcade/PCB machines with a CD-ROM or laserdisc in their ROM manifest
laserdisc games	Machines that use laserdisc media
cd-based	Arcade/PCB machines with CD-ROM hardware $\triangle$ Hardware filter — NOT the same as 'cdroms' which filters softlist CD media
samples / no samples	Show or hide games requiring external sound sample files
screenless	Machines with no display output
1 screen / one screen	Machines with exactly one display
2 screens / multi screen	Machines with two or more displays
3 screens / 4 screens	Machines with three or four displays
no protection	Hide games with emulated protection schemes
not bootleg / no bootlegs	Hide bootleg versions
cocktail / cocktail games	Machines supporting cocktail / flip-screen mode (HasCocktailDip = true)
no cocktail / upright only	Machines that do NOT support cocktail mode (HasCocktailDip = false)

cpu:z80	Machines using a specific CPU chip. Substring match on the chip list — cpu:z80 , cpu:68000 , cpu:arm . Multi-word names work: cpu:zilog z80 . Machines only — not available for softlist items.
sound:ym2151	Machines using a specific sound chip. Substring match on the audio chip list — sound:ym2151 , sound:oki , sound:pokey . Machines only — not available for softlist items.

CONTROLS

lightgun / lightgun games / light gun	Show lightgun games
no lightgun / no light gun	Hide lightgun games
trackball / dial / paddle	Games using these specific analog controls
dual stick / dual joystick	Games using a dual joystick setup
joystick	Games using any joystick input
2-way / 4-way / 8-way	Games with specific joystick directions
spinner / spinner games	Games using a spinner (rotary) control
mouse / mouse games	Games using a mouse or trackpad control
mahjong / mahjong games	Mahjong games (mahjong-keyboard input)
keyboard games	Games using a keyboard as primary input
keypad	Games using a numeric keypad
gambling games	Games using gambling-style input hardware

no buttons / 0 buttons	Games with no buttons
1 button / one button	Games with exactly 1 button
2 buttons ... 6 buttons	Games with 2 to 6 buttons (word forms also work: 'six buttons')
1 player ... 4 player	Filter by player count
multiplayer / co-op / 2p	Games supporting 2 or more players (also: 3p, 4p)
1p / singleplayer	Single-player games only

MEDIA TYPE (SOFTWARE LIST ITEMS)

These filter software list items by physical media type. For arcade CD hardware, use cd-based or chd games instead.

cartridges / cart games / modules	Cartridge-based software (media:cartridge)
cassettes / tape games	Cassette tape software (media:cassette)
floppies / floppy disks / diskettes	Floppy disk software (media:floppy)
cdroms / discs / disc games / cd-rom games / optical discs	CD-ROM software (media:cdrom)
hard disks / hard disk / hdds / hdd / hdd games	Hard disk software (media:harddisk)
memory cards / memory card / memcards / memcard	Memory card software (media:memcard)

MACHINE TYPE

Natural language phrases for machine type categories (requires history.xml in Art Paths). The machinetype: operator (aliases: mtype:, mt:) performs a substring match against the machine-type descriptor extracted from

history.xml. Some NLP keywords route via genre: (catlist.ini) instead where no history.xml descriptor exists — noted below.

fruit machines	Fruit machines / AWP (machinetype:fruit machine)
slot machines / slots	Video and mechanical slot machines (machinetype:slot machine)
pinball	Pinball machines (machinetype:pinball)
electro-mechanical / em games	Electro-mechanical games — routes via genre:Electromechanical (catlist.ini, not history.xml)
medal games	Medal / prize games (machinetype:medal)
home computers / home consoles	Home computers or consoles (machinetype:computer / machinetype:console)
handheld consoles / handheld games	Handheld gaming systems (machinetype:handheld)
calculators	Electronic calculators (machinetype:calculator)
chess games / chess computers	Chess computers and boards (machinetype:chess board)
musical instruments	Electronic musical instruments (machinetype:musical instrument)
tabletop games / tabletop	Tabletop game units (machinetype:tabletop)
jukeboxes / jukebox	Jukebox machines (machinetype:jukebox)
photo booths / photo booth	Photo booth machines (machinetype:photo booth)
redemption games	Redemption / prize games (machinetype:redemption)

dart games / dart game	Electronic dart games (machinetype:dart game)
bingo games	Bingo machines (machinetype:bingo)
kiddie rides	Kiddie ride machines (machinetype:kiddie ride)
bowlers / bowler	Bowling alley machines (machinetype:bowler)
gun games / gun game	Video gun games (machinetype:gun game)
quiz games / quiz machines	Quiz machines — routed via genre:Quiz (catlist.ini, not history.xml)

SOFTLIST SUPPORT STATUS (3-WAY)

Filter software list items by their MAME support level using supportstatus: (alias: ss:). Returns no results for arcade machines — combine with a system name for best results, e.g. snes supportstatus:yes.

supportstatus:yes / ss:yes	Fully supported softlist items (also: supported games, fully supported)
supportstatus:partial / ss:partial	Partially supported items (also: partially supported, partial support)
supportstatus:no / ss:no	Unsupported items (also: unsupported games, not supported)

COMPOUND NO-X AND RUN-TOGETHER FORMS

Most two-word “no X” and “non-X” phrases also work without the space or hyphen. These are fully equivalent to their spaced forms.

noclones	Same as no clones
nobootleg / nobootlegs	Same as no bootlegs
nomechanical / nonmechanical	Same as not mechanical

noprotection	Same as no protection
noworking	Same as nonworking
noarcade	Same as non-arcade (IsArcade = false)
nosamples	Same as no samples
nolightgun	Same as no lightgun
nodevices	Same as no devices
nosoftlist / nosoftware / nomedia	Same as machines only
nobios / nonbios	Same as no bios
coinop / coin op	Same as arcade only
coop	Same as co-op (multiplayer)
vertical only / horizontal only	Same as vertical / horizontal
noX (general pattern)	Most 'no X' and 'non-X' phrases support a compact no-space form. If 'no clones' works, 'noclones' works too.

DATE & DECADE FILTERING

Natural language date expressions are parsed automatically.

1982	Games from exactly 1982
79-82 / 1979-1983	Year range (2-digit or 4-digit, or mixed: 1999-02)

after 1990	Games released after 1990
before 1985 / pre-1985	Games released before 1985
made in 1982	Games from exactly 1982 (natural sentence form)
from 1980 to 1985	Inclusive year range
between 1991 and 1995	Inclusive year range (advanced syntax form)
80s games / in the 90s	Full decade (1980–1989, 1990–1999, etc.)
early 90s	First half of decade (1990–1993)
late 80s	Last third of decade (1987–1989)
00s games / in the 2010s	Works for any decade including 2000s and 2010s
mid 80s	Middle of a decade (1984–1986); works for any decade: mid 70s, mid 90s, etc.
since 1990	Open-ended: 1990 and later (Year >= 1990). Also: from 1990
through 1990	Open-ended: 1990 and earlier (Year <= 1990)

FIELD-SPECIFIC SEARCHES

Target a specific data field using field:value syntax. Colon (:) is equivalent to equals (=) and performs a contains-match on string fields.

year:1994	Games from 1994
manufacturer:capcom	Capcom games (contains match)

name:dkong	Games whose internal name contains 'dkong'
description:puzzle	Games with 'puzzle' in the title/description
cloneof:pacman	Clones of Pac-Man
sourcefile:cps2.cpp	Games on the CPS-2 hardware driver (aliases: hardware:cps2 or driver:cps2)
media:cartridge	Cartridge software (softlist items) (values: cartridge, cassette, floppy, cdrom, harddisk, memcard, rom)
DriverStatus:imperfect	Games with imperfect driver status
workingstatus:working / :imperfect / :notworking	3-way machine driver status filter (alias: ws:). Machines only. Finer-grained than IsWorking: which treats good and imperfect as equivalent
supportstatus:yes / :partial / :no	3-way softlist support level filter (alias: ss:). Softlist items only. Combine with a system: snes ss:yes
machinetype:fruit machine	Substring match against machine-type descriptor from history.xml (aliases: mtype:, mt:). Examples: mtype:computer, mt:pinball
mameversion:0.285 / mv:0.285	Exact match on MAME first-appearance version from MAMEInfo.dat. Also: version 285 (natural form — integer only)
history:scramble / hist:scramble	Search History.xml descriptive text for the given term
genre:Shooter / gg:Fighter	Substring match against CatList.ini category. Catches all subtypes automatically
HasLightgun:true	Direct boolean field access
IsArcade:false	Non-arcade machines

IsWorking:true	Working games (direct field form)
IsFavorite:true	Your favorite games (direct field form)
UsesSamples:true	Games requiring sample files
CloneOf:""	Parent games (empty CloneOf = not a clone)

Bare-word DAT operators: the colon is optional for DAT prefix fields. genre fighting, history pacman, workingstatus imperfect, machinetype pinball, and supportstatus partial all work identically to their colon forms. Multi-word values are captured correctly: machinetype:quiz machine finds quiz-type machines without 'machine' leaking into text search.

GENRE ALIASES

Genre keywords translate to genre: queries against catlist.ini (requires CatList.ini configured in Art Paths). Substring matching is used, so genre:Shooter catches all subtypes — Flying Vertical, Flying Horizontal, Run and Gun, etc. These aliases combine freely with other filters: shmups vertical working, brawlers noclones since 1990.

shmups / shoot em up / shooters	Shoot-'em-ups and all shooter subtypes (genre:Shooter via catlist.ini)
fighters / fighting games / vs fighters	All fighting game subtypes (genre:Fighter)
platformers / platform games / run and jump	Platform / jump-and-run games (genre:Platform)
brawlers / beat em ups / belt scrollers	Beat-'em-up / brawler games (genre:Beat)
racing / racing games / driving games	Racing and driving games (genre:Racing)
puzzle / puzzle games / puzzlers	Puzzle games (genre:Puzzle)
maze / maze games	Maze games (genre:Maze)

rpg / role playing	Role-playing games (genre:RPG)
sports / sports games	Sports games (genre:Sports)
adventure / adventure games	Adventure games (genre:Adventure)
casino / casino games	Casino games (genre:Casino)
baseball / basketball / soccer / boxing	Sport-specific catlist subgenres (e.g. genre:Baseball, genre:Boxing)

DIAGNOSTIC & QUALITY FILTERS

These filters target ROM dump quality and diagnostic cross-queries. Bad dump and no dump data are populated from the MAME XML at build time. Prototype detection uses a description substring match.

bad dump / bad dumps / bad roms	Machines with at least one known bad ROM dump (HasBadDumps = true)
no dump / nodump / missing dump	Machines with at least one ROM dump not yet found (HasNoDumps = true)
clean roms / good dumps / verified roms	No bad or missing dumps (HasBadDumps = false AND HasNoDumps = false)
prototype / prototypes / proto	Prototype and pre-release versions (description contains 'proto')
no prototypes / no proto	Exclude all prototype / pre-release entries
working clones of nonworking parents	Working clones whose parent set is not working (diagnostic cross-query)
nonworking parents with working clones	Non-working parents that have at least one working clone
working bios / nonworking bios	BIOS entries filtered by working status

working mechanical clones	Working clone entries that are also mechanical machines
nonworking mechanical parents	Non-working parent entries that are mechanical machines

FIELD NEGATION (!)

Prefix the value with ! to exclude matches.

manufacturer:!capcom	Every manufacturer EXCEPT Capcom
year:!1994	Every year EXCEPT 1994

ADVANCED OPERATORS

For precise control, use the full SQL-like operator syntax. These can be mixed freely with keyword modifiers.

=	Equality	manufacturer = "Capcom"
!=	Inequality	manufacturer != "Sega"
> >=	Greater than / or equal	year > 1990
< <=	Less than / or equal	year <= 1985
LIKE	Substring match (% is wildcard)	description LIKE "%Fighter%"
NOT LIKE	Substring exclusion	description NOT LIKE "%bootleg%"
BETWEEN	Inclusive numeric range	year BETWEEN 1990 AND 1992
IN	Matches any value in a list	manufacturer IN ("Taito", "Irem")
AND	Both conditions must be true	working AND year > 1990

OR	Either condition can be true manufacturer = "Sega" OR manufacturer = "Namco"	
NOT	Negates a condition	NOT (manufacturer = "Capcom")
()	Groups expressions	(manufacturer = "Sega" OR manufacturer = "Namco")

PUTTING IT ALL TOGETHER

Any combination of the above modes works in a single query. Keywords are translated to advanced syntax automatically before the filter runs.

Working, parent-only Midway games:

midway working only parents only

Capcom games from the late 90s, hiding Street Fighter:

capcom late 90s description NOT LIKE "%Street Fighter%"

All Sega or Namco games after 1990:

year > 1990 AND (manufacturer = "Sega" OR manufacturer = "Namco")

Parent puzzle games from 1995 or later:

Year >= "1995" noclones description LIKE "%puzzle%"

NES cartridge games (software list):

nes cartridges

Working 6-button arcade games, no clones, after 1991:

6 buttons working only noclones year > 1991 arcade only

Diagnostic: find playable versions of broken sets:

working clones of nonworking parents

All imperfect Sega games from the 90s:

sega workingstatus:imperfect 90s

Supported SNES cartridges only:

snest cartridges supportstatus:yes

Fruit machines added in MAME 0.196 or later:

fruit machines mameversion >= 0.196

APPENDIX 2 - USER REFERENCE

ARCHITECTURE

Application Architecture Summary

The IV/Play application is a high-performance MAME frontend for Windows. Its architecture is heavily optimized for fast startup, responsive filtering, and efficient rendering of large game lists by leveraging a deferred asset loading model, a DirectX-based rendering engine, a multi-layered caching strategy, and in-memory data processing.

Core Application Flow

The application's lifecycle begins in Program.cs, which ensures only a single instance is running from the same directory and handles the initial detection of the MAME executable if it's not configured.

The `MainForm` class serves as the central orchestrator. It features two distinct startup paths to maximize performance:

- **Cold Start (First Run / Database Rebuild):** On its first launch or after a MAME update is detected, the application performs a one-time data build. A splash screen is displayed while the `XmlParser` processes MAME's entire output and the `DatabaseManager` builds a local SQLite database. This is the longest startup path but is only required once per MAME version.
- **Warm Start ("Perfect Paint" Path):** This is the normal startup path, optimized to render an interactive UI that is visually complete — fully populated, accurately filtered, and ready for input — on the first frame. Rather than relying on a binary cache that could drift out of sync, the application performs a live ADO.NET query against the flattened MachinesSummary table, loading approximately 42,000 records in well under

half a second. A String Interning (Deduplication) pass runs during load, reducing memory pressure by collapsing repetitive strings into shared references. The result is a warm start that consistently reaches an interactive UI in under one second with no ghost entries, no placeholder icons, and no catch-up corrections after the fact. Heavier background tasks — icon atlas assembly, softlist index load, archive indexing — are deferred to a background thread and complete within roughly 400ms after the UI is already interactive, without blocking any user input.

Data Management and Persistence

IV/Play's data strategy is robust, designed to transform MAME's raw XML output into a queryable format.

- **Data Source:** The primary source of game information is the XML output from the `mame.exe -listxml` command.
- **Parsing:** The `XmlParser` class launches a MAME process and deserializes the resulting XML stream directly into a list of Machine objects. This process populates complex, nested properties for each machine, such as CPU, sound, and display information.
- **Persistence:** A `DatabaseManager` using raw ADO.NET (`SqlConnection` / `SqlCommand`) and SQLite stores the processed Machine data in a local database file (`IV-Play.db`). Entity Framework Core has been fully retired. The database is completely rebuilt if the MAME executable has been updated or if the application's database schema has changed, ensuring data consistency.
- **In-Memory Processing:** For maximum filtering speed, the entire ``Machines`` table is loaded from the database into a static list in memory (``DatabaseManager.AllMachines``) at startup. This is a key component of the 'Fast Path' that enables a sub-500ms time-to-interactive on warm starts.

Rendering and UI

The user interface is built for performance, moving beyond standard Windows Forms controls for its core component.

- **DirectX Engine:** The `GameList` control is a custom `UserControl` that bypasses standard GDI+ rendering. It uses the Vortice library to interface directly with DirectX (`Direct3D11` and `Direct2D`), enabling smooth scrolling and high-quality image scaling, transparency, and effects like scanlines.
- **Modular Control:** The `GameList` logic is separated into partial classes for drawing (`GameList.Drawing.cs`), user input (`GameList.Input.cs`), and navigation/data handling (`GameList.Navigation.cs`), keeping the code organized.
- **UI Management:** The `MainForm` manages the `GameList` control and auxiliary windows, such as the `ConfigForm` for settings and the `FilterDialog` for search queries.

Multi-Layered Caching Strategy

To achieve its high performance and rapid startup, the application employs several layers of caching.

- **Icon Atlas Cache:** `AssetCache` generates a single large PNG image (a "texture atlas") containing all game icons. This dramatically improves rendering performance by minimizing the number of individual images the GPU needs to manage. The atlas is cached to disk and validated using a hash of the source icon files, so it's only rebuilt when icons change. During a warm start, placeholder icons are rendered instantly while the full atlas is loaded from this cache on a background thread.
- **GameList Cache:** For the most common startup scenario (no active filters **or custom game list**), `GameListCache` saves the fully sorted and processed game lists to a compact binary file. Loading from this cache is significantly faster than querying the database and re-sorting thousands of entries.

- **Artwork Cache:** ArtCache implements a memory-based Least Recently Used (LRU) cache for game artwork like snapshots and flyers. This reduces disk reads for frequently viewed art. It uses a background ArtLoader to fetch images from disk asynchronously without blocking the UI.
- **DAT File Cache:** The InfoParser, used for history.xml and mameinfo.dat, creates a JSON-based cache of the parsed data. This cache is validated against the source file's timestamp, preventing redundant parsing of these large text files on subsequent runs.

Filtering and Search

The application features a sophisticated filtering system that allows for both simple and complex queries.

- **Natural Language Parser:** NaturalLanguageParser translates intuitive, human-readable queries (e.g., capcom fighting 1994) into the application's strict, SQL-like advanced filter syntax (e.g., manufacturer LIKE "%capcom%" AND description LIKE "%fighting%" AND year = "1994").
- **Advanced Filtering:** This advanced syntax supports a wide range of operators (=, >, LIKE, NOT, OR, AND, BETWEEN), parentheses for grouping, and specific keywords for fields like year, manufacturer, and driver.
- **In-Memory Execution:** The filter query is converted into a LINQ expression and executed directly against the in-memory list of all games (DatabaseManager.AllMachines), resulting in near-instantaneous filtering.

Configuration and Diagnostics

IV/Play includes tools for configuration and troubleshooting.

- **Settings Management:** A central, static SettingsManager class manages all user-configurable options. Settings are persisted to a text-based IV-Play.cfg file. The manager also handles the auto-detection of art folders and DAT files based on the MAME executable's location.
- **Diagnostics:** The application integrates a buffered Logger to record detailed events with minimal performance overhead by flushing writes to disk periodically rather than instantly. A StartupProfiler is also used to time key operations during the launch sequence, helping to identify and diagnose performance issues. It now reports both the primary "time-to-interactive" metric and a separate set of "Deferred Tasks (Post-UI)" timings in the log file.

APPENDIX 3 - DEEP TECHNICAL REFERENCE

This appendix is the implementation-level complement to Appendix 2. Where Appendix 2 describes the shape of IV/Play—its startup model, rendering pipeline, data flow, caching layers, filter engine, and diagnostics—this appendix describes how those systems are built and why they work the way they do. It is an evergreen reference: updated when the implementation changes, never allowed to accumulate history.

UNIFIED ARCHITECTURE

IV/Play is comprised of three core components: the Frontend, the Backend, and the SQLite Database. Each has a strict responsibility boundary that must be maintained to preserve performance and correctness.

- **Frontend (UI):** Pure WinForms shell. The core GameList control bypasses standard GDI+ rendering entirely and uses a custom engine built on DirectX (Direct3D11 and Direct2D) via the Vortice library. No GDI or GDI+ anywhere in the rendering path.

- **Backend (Data Layer):** Responsible for XML parsing, database management, cache orchestration, and filter execution. Uses raw ADO.NET (SqlConnection / SqlCommand) throughout—for schema creation, cold-start bulk inserts, softlist and DAT cache writes, and all warm-start reads. All connections are opened via OpenRuntimeConnection(), which applies WAL mode and foreign-key enforcement uniformly. Entity Framework Core has been retired.
- **SQLite Database (IV-Play.db):** Central repository for all game metadata. Two tables drive the application: Machines (full XML-derived records) and MachinesSummary (flattened, scroll-optimized, defined by MachineSummarySchema.Columns). For maximum filtering speed, the entire MachinesSummary table is loaded into the static in-memory list DatabaseManager.AllMachines at startup. All filter execution runs against this list—no database round-trips occur during user interaction.

RENDERING ENGINE

All drawing operations are performed by the GPU via DirectX 11 / Direct2D. The pipeline is built on the Vortice.Windows library and enforces strict no-allocation rules in the hot render path.

- **Pipeline components:** ID3D11Device and swap chain for 3D surface management. ID2D1DeviceContext for 2D drawing (text, vectors, borders, icons, art). Custom sprite batch for icon draw calls pulled from the GPU-resident atlas. IDWriteFactory / DirectWrite for all text rendering. High-DPI scaling matrix applied globally. Frame scheduler with vsync.
- **Layout geometry:** Fixed 60/40 split—the game list occupies the left 60% of the control width, the art panel the right 40%. Calculated in Gamelist.Drawing.cs as artZoneWidthInDips = widthInDips * 0.40f. Art panel bounds are stored in _imageArea (SysRectangleF). In Grid mode the grid occupies exactly the space the list would. In No-Art mode _imageArea is set to SysRectangleF.Empty and the full width is used.
- **DPI scaling:** GetDpiScale() returns the current scale factor (1.0 at 96 DPI, 1.5 at 144 DPI, 2.0 at 192 DPI). All layout calculations are performed in device-independent pixels (DIPs) and converted to physical pixels for DirectX calls. Multi-monitor DPI transitions are guarded by _atlasDpiScale, _dpiChanging, _shownComplete, and _renderSuspended flags in MainForm and GameList respectively.
- **Device-lost recovery:** Render entry is gated by a RenderState enum (Active / Resizing / Recovering). A _drawBatchOpen flag detects any stale BeginDraw left open by an earlier partial frame and forces recovery before a new draw begins. On confirmed device loss (D2DERR_RECREATE_TARGET), ForceDeviceRecovery() tears down and recreates the D2D context, fires DeviceLostRecovered on the UI thread, and triggers MainForm to rebuild the asset cache, re-upload the atlas, and reload art for the current selection. _isDisposed is reset on recovery so the next render cycle starts clean.
- **Art scaling:** DirectX High Quality Bicubic interpolation by default; nearest-neighbor available via art_scaling_nearest setting for a pixelated retro look. Scanline overlay (scanlines_enabled) is suppressed automatically for vector games and non-MAME-generated art types such as flyers, marquees, and PCBs.

GAMELIST PARTIAL CLASS STRUCTURE

The GameList control's logic is divided across nine partial class files, each with a strict responsibility boundary that must not be blurred:

- **Gamelist.cs** — Core state. Owns _mainGameList, _favoriteGameList, _activeGameList, _selectedGame, _filter, _dataLoaded, and _currentSystemProvider.
- **Gamelist.Drawing.cs** — All DirectX rendering. No GDI here or anywhere else.

- **Gamelist.Input.cs** — Keyboard and mouse handling.
- **Gamelist.Navigation.cs** — Scroll position, focus management, back/forward navigation stack.
- **Gamelist.Sorting.cs** — Sort logic for all view modes including Flat View global sort.
- **Gamelist.Predicates.cs** — Static filter predicate helpers such as `IsArcadeVideoGame()` and `MachineHasAtLeastScreens()`, used for conditions that cannot be expressed in pure SQL-like field comparison syntax.
- **Gamelist.DatOverlay.cs** — Tilde peek overlay rendering and sizing.
- **Gamelist.HistoryHighlight.cs** — DAT search term highlighting. Extracts active search terms from history; `mi:`, `mv:`, and plain-text filter components and highlights matching spans in the DAT overlay text area.
- **Gamelist.BVT.cs** — Build Verification Test harness (F10). Must remain synchronized with any changes to `GameList` interaction, navigation, or rendering logic.

DATA LAYER

RAW ADO.NET (ADO-ONLY)

The data layer uses raw ADO.NET exclusively. All database access—schema creation, cold-start bulk inserts to the `Machines` table, softlist and DAT cache writes, and all warm-start reads from `MachinesSummary`—flows through `SqlConnection` and `SqlCommand` via a private `OpenRuntimeConnection()` helper. This helper applies a single WAL-mode/NORMAL-synchronous/foreign-keys PRAGMA batch on every connection open, ensuring all connections are identically configured. Entity Framework Core has been fully retired; the `CompiledModels/` directory, the three EF context classes, and both `Microsoft.EntityFrameworkCore.*` NuGet packages no longer exist in the project.

MACHINESSUMMARY TABLE

`MachinesSummary` is the flattened, scroll-optimized projection of the `Machines` table. It is written during cold start via raw ADO.NET bulk insert and read at every warm start. Schema changes to this table require exactly one edit: the `MachineSummarySchema.Columns` descriptor array in `Data/Models/MachineSummarySchema.cs`. This array is the single source of truth that drives the DDL (`CREATE TABLE`), the `INSERT-SELECT` projection, the `SELECT` column list, and the reader index for `LoadAllMachinesIntoMemoryAsync()`. Previously, this required simultaneous updates to three separate sites in `DatabaseManager.cs`; violating that discipline caused a silent `SQLiteException` fallback. The schema descriptor eliminates this class of bug entirely.

`LoadAllMachinesIntoMemoryAsync()` returns a typed `CacheResult<List<MachineSummaryDto>>`. If the warm-start fast path cannot produce a Fresh result, it returns `CacheResult.Stale` and all callers log explicitly—there is no silent fallback to full hydration.

CACHE RESULT PROVENANCE

Cache-style operations across the application return `CacheResult<T>`—a zero-heap-allocation record struct carrying the data and a `CacheQuality` enum (`Fresh` / `Stale` / `Partial` / `Missing` / `Invalid` / `Failed`). Callers are required to inspect quality and log when the result is not `Fresh`; silent fallbacks are prohibited by convention. Affected subsystems: `SoftlistParser.ParseAll()`, `AssetCache.LoadFromCacheAsync()`, and

LoadAllMachinesIntoMemoryAsync(). DatInfoParser intentionally does not use CacheResult—its single call site has adequate inline discipline and the type would over-engineer it.

ATOMIC WRITES

All persistent state writes follow the write-to-temp-then-rename pattern: data is written to a .tmp sibling on the same NTFS volume, then promoted atomically via File.Move(overwrite: true). A crash anywhere in the write window leaves either the previous file intact or the new file complete—never a partial write. This pattern is enforced in three subsystems:

- **SettingsManager:** IV-Play.cfg is written to a .tmp file and renamed.
- **AssetCache:** The three-file atlas group (.bin / .map.json / .hash) is staged as three .tmp files, renamed in order, with the hash file written last as a seal—a missing or mismatched hash on load forces a full atlas rebuild.
- **DatabaseManager:** The database is rebuilt to IV-Play.db.new, then moved to the live path; the metadata seal is written to a .tmp and moved last. A crash anywhere in this window forces a rebuild on next launch rather than silently loading a corrupt file.

STRING INTERNING

During LoadAllMachinesIntoMemoryAsync(), a local Dictionary<string, string> string pool deduplicates all string values read from MachinesSummary. Without interning, the load process creates approximately 250,000 duplicate string objects—one per field per machine—for high-cardinality columns like Manufacturer and Year. Interning collapses these to approximately 14,000 unique references, saving hundreds of megabytes of RAM and eliminating a GC pressure spike that previously manifested as a UI stutter 5–10 seconds after launch.

COMMANDS TABLE

The Commands table is populated during cold start from mame.exe -showusage. At warm start, LoadMameCommandsAsync() reads this table into DatabaseManager.MameCommands (a Dictionary<string, string>), which drives the command-line override autocomplete in ConfigForm via the ComboGhostPainter NativeWindow subclass. This step is explicitly non-fatal—if the table is absent in an older database, loading proceeds normally and autocomplete is simply unavailable until the next cold start.

MULTI-LAYERED CACHING

- **Icon Atlas Cache (AssetCache):** All game icons are combined at cold start into a single PNG texture atlas (IV-Play.icon.atlas.png). Validity is checked on each warm start via an MD5 hash of the source icon directory (IV-Play.icon.atlas.hash). On a valid warm start, the atlas is loaded once to GPU memory on a background thread while placeholder icons render immediately. Supported sources: individual .ico files, icons.zip, or non-solid icons.7z.
- **Softlist Index Cache:** The ~140,000-item software list index is stored in a custom dictionary-compressed binary format (IV-Play.SoftlistIndex.cache). Unique strings are written into a header table and referenced by integer index, reducing file size and load times by approximately 25% compared to naive serialization. Loaded on a background thread after the UI is interactive.

- **Artwork Cache (ArtCache):** Memory-based LRU cache for snapshots, flyers, and other art types. A background ArtLoader fetches images asynchronously so the UI thread is never blocked. Supports both loose files and .zip / non-solid .7z archives. Solid .7z archives are detected and warned against on startup due to their incompatibility with random-access performance.
- **DAT File Cache:** History.xml and MAMEInfo.dat are parsed once on first use into binary cache files (IV-Play.history.cache, IV-Play.mameinfo.cache), validated against source file timestamps on all subsequent launches. In version 2.6, these caches are promoted from passive peek-overlay sources to first-class participants in the filter engine via MetadataHydrator. Cold start parses 173,539 History.xml entries (~998ms) and 14,674 MAMEInfo.dat entries (~784ms); all subsequent warm launches load from cache in milliseconds.

FILTER ENGINE

PIPELINE

Filter execution follows a fixed pipeline: Tokenize → Normalize → NLP translation → Build predicate → Apply to DatabaseManager.AllMachines (in-memory LINQ). NaturalLanguageParser.cs translates human-readable queries into SQL-like advanced filter syntax. FilterExpressionParser.cs tokenizes the resulting string, builds a recursive-descent expression tree (ParseOrExpression → ParseAndExpression → ParseComparison), and compiles it into a LINQ lambda. The compiled predicate is applied to the in-memory list; no database round-trips occur.

DAT INTEGRATION AND METADATAHYDRATOR

History.xml, MAMEInfo.dat, and CatList.ini participate in the filter engine as first-class data sources. MetadataHydrator is a lazy, one-shot-per-session static class that hydrates runtime-only Machine fields from these sources the first time a DAT-based filter operator is typed. Once triggered it hydrates all ~42,000 machines and sets a completion flag; subsequent filter executions pay no additional hydration cost. ResetHydrationState() is called when the MAME executable changes to force re-hydration against new DAT files.

DAT filter operators: history: / hist: searches History.xml descriptive text; genre: / gg: queries CatList.ini categories and subgenres; mameversion: / mv: queries MAMEInfo.dat first-appearance version data; machinetype: / mtype: / mt: performs a substring match against the machine-type descriptor from History.xml. All four participate in the natural language layer. workingstatus: / ws: provides three-way driver status filtering (working / imperfect / notworking). supportstatus: / ss: filters softlist items by MAME support level (yes / partial / no).

AUTOCOMPLETE

The filter bar (Ctrl-F) uses GhostTextBox, a TextBox subclass that renders VSCode-style inline ghost text via WndProc/Wm_Paint interception. The ConfigForm command-line override field uses ComboGhostPainter, a NativeWindow subclass that attaches to the internal edit control of a DropDown ComboBox and renders ghost completion text against DatabaseManager.MameCommands. WinForms TextBox controls do not participate in the standard paint pipeline, making WndProc interception the only reliable approach for inline ghost text in this framework.

AVAILABLE GAMELIST PIPELINE

AvailableListProvider.BuildAndSave() performs a name-only, non-recursive audit of all rompath entries in MAME.ini, matching archive names against DatabaseManager.AllMachines. Results are serialized to IV-Play.available.bin. This file is loaded at startup if present and drives the Available and Missing gamelist domains in the custom gamelist cycle. The scan is triggered by F12 and takes a few seconds on large ROM sets; the application auto-switches to the Available view on completion. ROM-less machines such as Pong are intentionally excluded—the intent is to show only machines with a physical ROM archive present. The cache file is deleted on F8 factory reset; if it is absent the Available and Missing entries appear in the cycle with a hint label prompting the user to press F12.

FLAT VIEW AND GLOBAL SORT

Flat View is a non-indented display mode in which clones are decoupled from their parents and favorites are not pinned at the top. It is the only view mode that supports global sort. The sort field (Description, Machine Name, Manufacturer, Year) is stored in the sort_flat_alphabetical setting and persists across restarts when Flat View is active; sort direction resets to its natural default on launch. Sort field cycles via Alt-S; sort direction via Shift-Alt-S. Gamelist.Sorting.cs owns the sort logic and must be updated if new sort fields are added.

MACHINE STATUS-AWARE ICON BORDERS

Icon borders are colored to reflect MAME driver status: working (green), imperfect (orange), not working (red). Colors are configurable per-status in the settings; the imperfect color defaults to a background-derived themed color to complement the dynamic theme system, and can be overridden to a fixed RGB value in F1. The border coloring extends to softlist media items. The implementation draws the status-colored border in Gamelist.Drawing.cs after the icon sprite is rendered, using Direct2D rectangle drawing with the status color applied as a stroke.

SETTINGS MANAGEMENT AND SNAPSHOT SYSTEM

SettingsManager is a static class whose public properties are decorated with [Category(...)] attributes for INI section grouping. Settings are persisted to the human-editable IV-Play.cfg file. Three-site discipline applies to any new setting: declaration, serialization, and all consumer sites must be updated together.

DumpSettingsSnapshot(label, SnapshotRuntimeContext, emitToLog) captures the full in-memory settings state at three points per session: application startup (“Startup” label), immediately before F1 opens (“Pre-F1”), and immediately after F1 closes (“Post-F1”). A delta marker (Δ) is appended to any field whose value changed from the previous snapshot. The most recent Post-F1 block is pink-highlighted in the F2 log overlay. The _lastSnapshotLines field stores the previous snapshot body for comparison; the Startup snapshot is never annotated as it establishes the baseline.

DIAGNOSTICS

- **Logger:** Structured producer/consumer logger. All call sites use a typed API — Logger.Info(LogCategory, message), Logger.Warn, Logger.Debug, Logger.Error — with an explicit subsystem category on every entry. The line format is: hh:mm:ss.fff tt [T:nn] [SEV] [CAT] message, where T:nn is the managed thread ID, SEV is one of VRB/DBG/INF/WRN/ERR, and CAT is one of 19 subsystem tags: PRG (startup), CFG (settings), DB (database), DAT (XML/DAT parsing), GUI (main form), ARC (archive provider), AST (icon

atlas), BKG (background manager), AVL (available list), MAM (MAME process), FAV (favorites), GEO (geometry), DPI (DPI scaling), MON (monitor enumeration), WIN (window state), SPL (splash), ART (art loading), AUD (auditor), GEN (general/unclassified). Entries are queued into a bounded BlockingCollection (capacity 2,000) and drained by a dedicated consumer thread running at BelowNormal priority, keeping the hot path allocation-free. If the queue is full, the caller falls back to a direct synchronous write rather than blocking or dropping. Error-level entries always pass regardless of MinimumLevel or CategoryFilter. Verbose logging is toggled via F1 (debug_mode); a per-category filter (debug_category_filter in the cfg, comma-separated category names) is available for targeted subsystem isolation. The log file rotates at 5 MB, keeping three generations (.1/.2/.3). FlushNow() provides a synchronous sentinel-based drain used by Stop() and crash handlers. In the F2 log overlay, each category is rendered in a distinct canonical color defined in Logger.CategoryColors, making subsystem activity immediately scannable. F2 state-dump sections (written by the Ctrl+Shift+0 state snapshot) are stripped from the log file on clean exit via RemoveF2DumpsFromFile(). Legacy call sites (WriteToLog, LogDebug) are retained as [Obsolete] and route to the typed API with LogCategory.GEN.

- **StartupProfiler:** Static class recording wall-clock timestamps for key startup events in two categories: the interactive-phase events that contribute to Time-to-Interactive (WallClockTime_ToInteractive), and deferred post-UI events added via AddDeferredEvent(). The full bar-chart report is written to the log and the Time-to-Interactive figure is surfaced on the F1 configuration screen.
- **Developer overlays:** F2 displays the IV-Play.log peek overlay with the Settings Snapshot delta system. F3 displays the IV-Play.cfg overlay. F7 shows the real-time performance dashboard (FPS, CPU memory, GPU memory, GC). F6 launches the asset auditor. F10 runs the BVT. All overlays share the same navigation model as the DAT peek (arrow keys, Page Up/Down, Home/End).

LIVE ROM MONITORING (FILEWATCHERSERVICE / CHANGEQUEUE / BATCHPROCESSOR)

Three classes cooperate to keep the Available gamelist current without user intervention: **ChangeQueue** is a thread-safe, HashSet-backed dedup queue. Duplicate paths received before a drain are collapsed to one entry. A “full rescan” sentinel (null path) can be raised and is sticky until the next drain, ensuring no change event is lost even during a buffer overflow. **FileWatcherService** wraps one FileSystemWatcher per ROM directory (watching *.zip and *.7z, top-level only) and a separate FSW for the directory containing favorites.ini (filtered to that filename). All events funnel into the shared ChangeQueue. FSW buffer overflow triggers EnqueueFullRescan so no change is silently dropped. **BatchProcessor** drains the queue on a 500ms timer tick with a 1,000ms stability window: if any enqueue occurred within the last second the tick is skipped, preventing a burst of FSW events (e.g. a large copy) from triggering many sequential rebuilds. A single rebuild fires approximately 1s after the last file settles. Batch routing: a null batch (full-rescan sentinel) or a ROM archive path triggers AvailableListProvider.BuildAndSave(); a favorites.ini path triggers DatabaseManager.SyncFavoritesAsync(). Start/stop is owned by MainForm; the processor does not auto-start. RomChangeProcessed and FavoritesChangeProcessed events notify callers on completion. The service is active during both warm and cold start paths; the log confirms 7 watcher instances (3 ROM dirs × 2 extensions + favorites.ini) at each session start.

EXTRACTED SERVICE CLASSES

Several responsibilities previously embedded in larger classes were extracted into standalone service classes during the May 2026 re-org:

- **InstanceLock** (extracted from Program.cs): Produces stable, path-derived names for the single-instance mutex and any future named pipes, so that multiple side-by-side installations each get their own lock. Naming strategy: SHA-256 of the normalized startup path, hex-encoded and lower-cased. GetMutexName(prefix) returns “Global\{prefix}-{fullHash}”; GetPipeName(prefix) returns the first 16 hex characters of the same hash. Hash is computed once and cached.
- **InternedStringPool** (extracted from DatabaseManager inline logic): Canonical string-interning helper shared across all subsystems that deduplicate high-cardinality strings at load time (machine names, manufacturers, genres, source files). Delegates to string.Intern(), placing values in the CLR’s permanent intern table for process lifetime. Null-safe: returns null when passed null. The warm-start profiler now reports StringPool size: 0, confirming the CLR intern table rather than a local dictionary is the active path.
- **LayoutMetrics** (Phase 3 DPI system): Centralized DPI-aware layout constants authored at a single DesignDpi of 144 DPI (the 4K/HiDPI baseline). At 144 DPI runtime, scale = 1.0 (pixel-perfect); at 96 DPI, scale = 0.667 and design values downscale to their 96-DPI equivalent. D2D properties return DIPs directly (design unit × toDip, where toDip = 96/144). Window geometry properties return 144-DPI pixels directly for callers that apply RuntimeDpi/DesignDpi themselves. Initialize(runtimeDpi) is called once after Application.SetHighDpiMode; Update(newDpi) responds to WM_DPICHANGED.
- **HiscoreLoader** (new): Loads Hiscore.ini from the executable directory at startup. Format: romname TAB score TAB date (date optional). After loading, the file is rewritten sorted alphabetically by ROM name. Parsing is intentionally lenient: literal two-char escape “\t” is normalized to a real tab; fallbacks handle space-padded files and single-whitespace boundaries. Stub lines (ROM name only, no score) are preserved on save. Entries are available in HiscoreLoader.Entries (Dictionary keyed case-insensitively by ROM shortname). No hot-reload; loaded once at startup. The warm and cold logs confirm 212 entries, 0 stubs at startup.
- **Relocations**: GameListSearch was moved from Controls/ to Classes/ to align with its non-UI data-only role. SoftwareItem was moved from Classes/ to Data/Models/ to co-locate it with its peers. Neither relocation changes behavior.

NATIVE AOT RUNTIME MODEL

Beginning with version 2.6, IV/Play ships as a Native AOT (Ahead-of-Time compiled) executable. The .NET 10 runtime is embedded in the binary and is not a separate installation requirement. The distributable is larger as a result, but startup behavior is more consistent across machines and warm launches are smoother. AOT compilation required resolution of WinForms UIA reflection constraints. EF Core has since been fully retired; all database access is raw ADO.NET, which is AOT-safe by default. The S.cs helper class provides AOT-safe string obfuscation using Base64/XOR encoding; the encode method is present only in DEBUG builds.

CURRENT PERFORMANCE SNAPSHOT

The following metrics reflect the current release (v2.8.4, May 2026) on the reference development system. These figures are updated with each release; historical comparison data is in Appendix V.

Test configuration: i7-12700K, 64GB DDR4, NVMe SSD, 4K Primary (3840×2160, 144 DPI) + 1440p Secondary (2560×1440). IV/Play 2.8.4, MAME 0.287, 42,839 machines, 140,763 software list items, Native AOT (.NET 10).

Warm start milestones: [1] Initial I/O ~69ms → [2] UI Interactive 957ms → [3] Full Startup ~1.07s.

Interactive-phase breakdown (TTI 972ms): DB Load (EagerInit) 684ms • DB Load (main thread, overlapping) 502ms • Softlist Index Load 179ms • Settings Load 57ms • DirectX Init 42ms • Background Load 18ms. Note: FilterGamesAsync and Gamelist Filter (RAM) are no longer measured in the warm-start TTI critical path following the May 2026 re-org.

Deferred tasks (post-UI, ~408ms background): Atlas Valid+Load 246ms • Art Path/Archive Init 100ms • Softlist Index Load 179ms.

Cold start (one-time per MAME version): Atlas Build 3.65s • XML Parse & Process 2.52s • DB Write 1.11s • DAT Load History.xml 879ms • DAT Load MAMEInfo.dat 663ms • Internal Ready 10.79s • Launch-to-splash 193ms.

APPENDIX 4 - DEVELOPMENT HISTORY

This appendix is the engineering archaeology of IV/Play—the record of how the application got to where it is. It documents eleven months of modernization from June 2025 through May 2026, structured around the nine architectural arcs that define the system today. Unlike Appendices 2 and 3, this appendix is append-only: the history is not rewritten, only extended. New entries are added when a significant architectural arc completes.

FOUNDATION STABILIZATION & EARLY ARCHITECTURE (JUNE–JULY 2025)

The project enters source control on June 30, 2025, already mid-flight. The first weeks are chaotic: focus issues on launch, broken favorites navigation, icon regressions, scroll judder, properties panel failing to populate, softlist launch commandline malformed, and several sessions accidentally committing handoff summaries as commit messages. Despite the turbulence, the foundations land: SQLite replaces the old database format, MAME XML parsing is hardened, softlist scraping begins, the properties panel appears, and the startup pipeline gets its first performance pass. A binary DB cache is attempted and then abandoned due to ghost entries and GC spikes. String interning is introduced—the technique that survives to the present day.

RENDERING, DPI, AND SOFTLIST DEPTH (AUGUST–SEPTEMBER 2025)

This period is dominated by DPI battles and softlist ordering nightmares. The icon atlas becomes DPI-aware and the high-DPI scaling matrix is stabilized. Softlist parent/clone sorting repeatedly breaks: NTSC/PAL ordering takes multiple attempts, and region-variant orphans require hand-tuned fixes. Properties view expands with CPU, sound, and softlist metadata. The non-working overlay is fixed. Aspect ratio heuristics for snaps are introduced. A custom color picker for art borders and scrollbar thumbs debuts. The rendering pipeline becomes robust enough to support all later expansions.

DYNAMIC THEMES, SCROLLBAR, AND INPUT INFRASTRUCTURE (OCTOBER 2025)

October is dominated by the dynamic theme engine: five themes derived from background hue, ESC cancellation logic, brightness detection, theme randomizer, and conflicts between custom colors and theme overrides. The vertical scrollbar lands—a deceptively simple feature that triggers weeks of layout and DPI adjustments. ShortcutManager and FilterDialog autocomplete infrastructure appear, laying groundwork for the filter engine rewrite.

SOFTLISTS BECOME FIRST-CLASS CITIZENS (NOVEMBER 2025)

The first major architectural inflection point. 140,000 softlist items become filterable. The unified filter pipeline (BuildGameLists) is born. Arcade machines are sorted above ports. Archive support for .zip and .7z art assets lands. Multi-instance MAME launch (Shift-1 through 9) is added. The SQLite MachinesSummary summary table is introduced. The performance dashboard is rewritten multiple times. IV/Play becomes a mixed-domain frontend for the first time.

MIXED-VIEW FILTER HARDENING (DECEMBER 2025)

December is a war zone of filter bugs: dkong softlist produces inconsistent results, NTSC/PAL ordering breaks repeatedly, the Dendy ordering problem resurfaces, noclones interacts badly with softlist results, softlist favorites fail to appear after cold start. The app begins hiding itself on MAME launch. Raw ADO.NET bulk insert accelerates DB writes. The binary gamelist cache is retired in favor of a live ADO.NET query—the Perfect Paint architecture. A schema bug silently breaks the Fast Path by omitting the Features column from the MachinesSummary CREATE TABLE statement; the Fast Path falls back to full EF Core hydration on every warm start without surfacing a clear user-facing error. This bug persists until February 2026.

FAMILY GROUPING, PROPERTIES OVERHAUL, AND THE SCHEMA BUG FIX (JANUARY–FEBRUARY 2026)

January introduces family grouping—collapsing hundreds of softlist variants into drillable nodes. Expand/collapse logic breaks navigation repeatedly; ESC fails to restore the originating entry; the back-stack logic is rewritten. Version 2.4.12 ships. February uncovers the Features column schema bug: MachinesSummary CREATE TABLE was missing the Features column while both INSERT and SELECT referenced it, silently adding ~157ms to every warm start since December. Fixed in 2.4.14 by adding the missing column definition to DatabaseManager.cs. F8 factory reset could not fix it because the bug was in the code itself, not the data. Properties view gets a major overhaul: cocktail mode detection, bad/no-dump flags, working status aligned with MAME’s color coding. Performance dashboard switches to bar chart format only.

FILTER POLISH, AUTOCOMPLETE, DAT INTEGRATION (LATE FEBRUARY–EARLY MARCH 2026)

GhostTextBox (filter autocomplete) and ComboGhostPainter (command-line autocomplete) land. The Commands table from mame.exe -showusage drives command-line completion. History.xml, CatList.ini, and MAMEInfo.dat become fully filterable via MetadataHydrator. HiddenIniParser is introduced. Flat View with Global Sort (Alt-S) ships. Show Machines / Show Parents toggles are added. AvailableListProvider implements the F12 ROM scan. The Settings Snapshot delta marker system (DumpSettingsSnapshot, SnapshotRuntimeContext) is added. DAT search term highlighting (Gamelist.HistoryHighlight.cs) is implemented. DPI multi-monitor hardening adds _atlasDpiScale, _dpiChanging, _shownComplete, and _renderSuspended flags. Version 2.6 ships.

MIXED-VIEW UNIFICATION AND NATIVE AOT (MARCH 2026)

The second major architectural inflection point. Softlist predicate matching, softlist proxy construction, and unified filtering of machines plus softlists are completed. Correct bypass of softlist toggles when user intent is explicit. Correct merging into the unified list. Native AOT transition requires EF Core compiled model generation and resolution of a WinForms UIA reflection crash (DataGridViewRowHeaderCell parameterless constructor trimmed by the AOT linker). The FilterGamesAsync cost rises to ~331ms as a consequence of DAT data participating in the

filter predicate build pass across 42,700+ machines—an intentional architectural trade-off. Version 2.6.2 ships as the first Native AOT build.

FORENSIC AUDIT & STRUCTURAL HARDENING (APRIL 2026)

A multi-session forensic audit across five architectural arcs closed the primary classes of structural fragility that had accumulated since the 2.0 modernization.

Arc 1 — Atomic Writes. Three non-atomic write windows were closed: IV-Play.cfg (SettingsManager), the three-file icon atlas (AssetCache), and the database+metadata pair (DatabaseManager). All three now follow write-temp/atomic-rename on the same NTFS volume.

Arc 2 — Device-Lost State Machine. The rendering recovery path had three silent failure modes: `_isDisposed` was not reset on recovery, `D2DERR_RECREATE_TARGET` was not in the catch filter, and atlas/art/background were not re-uploaded after context recreation. All three are resolved. `RenderState` (Active/Resizing/Recovering) now gates `Render` entry, and a `_drawBatchOpen` flag detects stale `BeginDraw` before each frame.

Arc 3 — MachinesSummary Schema Discipline. The three-site manual synchronization contract (DDL / INSERT / SELECT reader) that had already caused one silent regression (the December 2025–February 2026 Features column omission) was replaced by `MachineSummarySchema.Columns`—a single descriptor array that drives all four sites. One edit adds a column correctly everywhere.

Arc 4 — Cache Result Provenance. `CacheResult<T>` (record struct, zero heap allocation) replaced null-checks and ad-hoc wasPartial tuples across `SoftlistParser`, `AssetCache`, and `DatabaseManager`. Callers are now required to log when quality is not Fresh.

EF Core Retirement. The EF Core / ADO.NET hybrid—the largest structural liability identified in the audit—was fully resolved: both `Microsoft.EntityFrameworkCore.*` packages removed, three EF context classes deleted, all four `using(context)` blocks in `DatabaseManager` converted to raw `SqlConnection` + `SqlCommand`, `CompiledModels/` (twelve generated files) deleted from the repository. The `OpenRuntimeConnection()` helper now enforces WAL mode and foreign keys on every connection uniformly.

Three-layer crash net (`AppDomain.UnhandledException`, `Application.ThreadException`, `MainForm` constructor try/catch writing timestamped crash logs to `%APPDATA%/IV-Play/crash_*.txt`) was confirmed present in source during the audit.

The audit established five architectural invariants that apply to all future work: atomic writes for all persistent state; no silent fallbacks (`CacheResult<T>` required); schema invariants in code (`MachineSummarySchema.Columns`); no EF Core; device-lost recovery via `RenderState` state machine.

STRUCTURAL RE-ORG AND SERVICE EXTRACTION (MAY 2026)

Following the April 2026 forensic audit, a structured architectural review identified several classes whose responsibilities had grown beyond their original scope. The May 2026 re-org extracted those responsibilities into purpose-built service classes and reorganized file placement to match logical ownership. The re-org produced a measurable performance improvement: `FilterGamesAsync` and `Gamelist Filter` (RAM) were moved out of the warm-start TTI critical path, reducing warm TTI from 972ms (v2.6.2) to 957ms (v2.8.4) despite 112 additional machines and a newer MAME version.

New services. The FileWatcherService / ChangeQueue / BatchProcessor triad implements live ROM availability monitoring without user intervention. FileWatcherService wraps one FSW per ROM directory and one for favorites.ini. ChangeQueue provides thread-safe dedup with a full-rescan sentinel. BatchProcessor drains the queue on a 500ms timer with a 1,000ms stability window and routes each batch to either AvailableListProvider (ROM changes) or DatabaseManager.SyncFavoritesAsync (favorites changes). HiscoreLoader introduces a Hiscore.ini subsystem: load once at startup, sort alphabetically by ROM name on write, lenient parsing of tab/space-separated formats, stub-line preservation.

Extracted helpers. InstanceLock extracted the SHA-256 path-derived mutex naming logic from Program.cs into a dedicated class, enabling future named-pipe usage with the same path identity. InternedStringPool extracted the CLR string.Intern() delegation from DatabaseManager's load-time inline code into a shared, null-safe wrapper available to all subsystems. LayoutMetrics (Phase 3) centralized all DPI-aware layout constants at a DesignDpi of 144, providing DIPs directly to D2D callers and eliminating scattered conversion arithmetic.

File relocations. GameListSearch moved from Controls/ to Classes/ (data class with no UI dependency). SoftwareItem moved from Classes/ to Data/Models/ (peers with other model types). CacheResult and MachineSummarySchema were already in their correct locations from the April audit.

KEY BUG ARCHIVE

The following table records significant bugs that shaped the architecture. It is a historical record, not a current issue tracker.

Category	Description	Resolution
Stability	Application crashed on exit after DirectX migration.	DirectX resources now disposed synchronously on the UI thread in the correct order.
Stability	NullReferenceException on startup after the God Object refactor.	DirectX initialization now tied to the OnHandleCreated event.
Performance	Scroll judder during fast navigation.	Entire rendering pipeline migrated from GDI+ to DirectX.
Performance	Multi-second bottleneck during type-to-search.	Search re-architected to query a lightweight in-memory list rather than full Machine objects.
Rendering	1-pixel transparent seam around art panel border on high-contrast backgrounds.	Anti-aliasing temporarily disabled for border draw calls in the DirectX renderer.
Navigation	"Stuck on S" — left-arrow navigation stuck or jumping erratically.	Letter-jump navigation logic fixed; sort key handling corrected.
Data	Hide Non-Working produced incorrect game counts vs. MAME.	Data model and parser corrected to align with MAME's internal logic (attributes vs. feature tags).

Category	Description	Resolution
Schema	Missing Features column in MachinesSummary CREATE TABLE silently forced full EF Core hydration fallback on every warm start from December 2025–February 2026, adding ~157ms.	Features TEXT column added to the CREATE TABLE statement in DatabaseManager.cs (v2.4.14). F8 reset could not fix it—the bug was in the code, not the data.
Software Lists	Entering a software list crashed due to duplicate names.	Software items de-duplicated by name before dictionary insertion.
Software Lists	Games within a software list failed to launch.	Argument construction logic corrected to generate the proper MAME command (e.g., a2600 -cart combat).
AOT	WinForms UIA reflection crash: DataGridViewRowHeaderCell parameterless constructor trimmed by the AOT linker.	Resolved via AOT-specific annotations. EF Core compiled model generation added for AOT-safe database access.

APPENDIX 5 - PERFORMANCE DELTA TIMELINE

This appendix is IV/Play’s longitudinal performance record. Every major changed release adds a new entry. Nothing is removed or rewritten. The intent is to track how the startup model evolves across MAME versions, hardware generations, and architectural changes. All measurements use the internal StartupProfiler instrumentation on the reference development system unless otherwise noted.

Reference hardware (all entries unless noted): i7-12700K, 64GB DDR4, NVMe SSD, 4K Primary (3840×2160, 144 DPI) + 1440p Secondary (2560×1440).

ARCHITECTURAL EVOLUTION SUMMARY

The Time-to-Interactive (TTI) on a warm start evolved as follows across the 2025–2026 modernization period:

Date / Milestone	TTI (Warm Start)	Key Architectural Change
July 1, 2025	~2.9s	Baseline before major refactoring.
July 5, 2025	~5.0s	Regression after initial un-optimized SQLite migration.
July 29, 2025	497ms	“Pure Speed” architecture: binary gamelist cache. Fast but produced ghost entries and GC spikes.
Sept 28, 2025	195ms	Peak raw speed. Relied on potentially stale cache; ~250K duplicate string objects; GC stutter 5–10s post-launch. Discarded.
December 2025	~1,040ms*	Perfect Paint architecture. Binary cache retired; live ADO.NET query. Fast Path broken by missing Features column—full EF

Date / Milestone	TTI (Warm Start)	Key Architectural Change
		Core hydration fallback on every warm start. *Measured reality, not documented target.
February 2026 (v2.4.14)	906ms	Fast Path restored (Features column fix). .NET 10. String interning active. 13% improvement over broken state.
March 2026 (v2.6.2)	972ms	Native AOT. DAT data in filter scope (+263ms FilterGamesAsync, intentional trade-off). Still sub-1-second Perfect Paint.
May 2026 (v2.8.4)	957ms	Structural re-org. FilterGamesAsync removed from TTI critical path. FileWatcherService/ChangeQueue/BatchProcessor triad. InstanceLock, InternedStringPool, LayoutMetrics Phase 3, HiscoreLoader extracted. File relocations (GameListSearch, SoftwareItem).

ENTRY 1: V2.4.14 — FEBRUARY 16, 2026

Software: IV/Play 2.4.14, MAME 0.285, 42,639 machines, 140,278 software list items, .NET 10.0 (JIT).

Key change: Fast Path ADO.NET loader restored. Missing Features column in MachinesSummary CREATE TABLE fixed. String interning active.

Warm start milestones: [1] Initial I/O 45ms → [2] UI Interactive 906ms → [3] Full Startup 1,340ms.

Interactive-phase breakdown (TTI 906ms): DB Load Summary Raw 341ms (39.6%) • FilterGamesAsync 68ms (7.9%) • Gamelist Filter RAM 50ms (5.8%) • DirectX Init 40ms (4.6%) • Show & Activate UI 12ms (1.4%) • Other/UI Load 350ms (40.7%).

Deferred tasks (post-UI, 434ms background): Atlas Valid+Load 244ms (56.2%) • Softlist Index Load 135ms (31.1%) • Art Path/Archive Init 63ms (14.5%) • Favorites Sync 7ms (1.6%).

Notes: DB load 71ms slower than the December 2025 design target of 270ms, attributable to data growth (MAME 0.285), string interning overhead, and real-world vs. controlled-test variance. The 906ms TTI represents a 13% improvement over the broken-Fast-Path state (1,040ms).

ENTRY 2: V2.6.2 — MARCH 17, 2026

Software: IV/Play 2.6.2, MAME 0.286, 42,727 machines, 140,360 software list items, Native AOT (.NET 10).

Key change: Native AOT. History.xml, MAMEInfo.dat, CatList.ini promoted to first-class filter participants via MetadataHydrator. FilterGamesAsync cost rises from 68ms to 331ms (intentional trade-off for substantially richer search capability).

Warm start milestones: [1] Initial I/O ~56ms → [2] UI Interactive 972ms → [3] Full Startup 1.38s.

Interactive-phase breakdown (TTI 972ms): DB Load Summary Raw 371ms (26.9%) • FilterGamesAsync 331ms (24.0%) • Gamelist Filter RAM 329ms (23.9%) • DirectX Init 109ms (7.9%) • Show & Activate UI 13ms (0.9%).

Deferred tasks (post-UI, ~408ms background): Softlist Index Load 204ms • Atlas Valid+Load 192ms • Art Path/Archive Init 106ms • Favorites Sync 13ms.

Cold start (one-time per MAME version): Atlas Build 3.86s • XML Parse & Process 2.58s • DB Write 1.48s • DAT Load History.xml 998ms • DAT Load MAMEInfo.dat 784ms. Total ~12.1s.

Delta vs. v2.4.14: DB Load +30ms (data growth) • FilterGamesAsync +263ms (DAT scope) • TTI +66ms • Total Launch +40ms. Perfect Paint guarantee maintained at sub-1-second TTI.

Notes:

ENTRY 3: V2.8.4 — MAY 10, 2026

Software: IV/Play 2.8.4, MAME 0.287, 42,839 machines, 140,763 software list items, Native AOT (.NET 10).

Key change: Structural re-org. FilterGamesAsync and Gamelist Filter (RAM) removed from warm-start TTI critical path. Live ROM monitoring triad (FileWatcherService / ChangeQueue / BatchProcessor) added. Extracted: InstanceLock, InternedStringPool, LayoutMetrics Phase 3 (DesignDpi=144), HiscoreLoader. File relocations: GameListSearch (Controls→Classes), SoftwareItem (Classes→Data/Models).

Warm start milestones: [1] Initial I/O ~69ms → [2] UI Interactive 957ms → [3] Full Startup ~1.07s.

Interactive-phase breakdown (TTI 957ms): DB Load (EagerInit, T:04) 684ms • DB Load (main thread, overlapping) 502ms • Softlist Index Load 179ms • Settings Load 57ms • DirectX Init 42ms • Background Load 18ms • GetMameInfo 2ms. FilterGamesAsync and Gamelist Filter (RAM) no longer appear in TTI scope.

Deferred tasks (post-UI, background): Atlas Valid+Load 246ms • Art Path/Archive Init 100ms • Softlist Index Load 179ms.

Cold start (one-time per MAME version): Atlas Build 3.65s • XML Parse & Process 2.52s • DB Write 1.11s • DAT Load History.xml 879ms • DAT Load MAMEInfo.dat 663ms • Internal Ready 10.79s • Launch-to-splash 193ms.

Delta vs. v2.6.2: DB Load -32ms (EagerInit overlap) • FilterGamesAsync 0ms in TTI (-331ms from critical path) • TTI -15ms • Cold Atlas Build -0.21s • Cold DB Write -0.37s • Cold DAT History -119ms • Cold DAT Mameinfo -121ms. Perfect Paint guarantee maintained. Machines +112. Software items +403.

Notes: The dual DB-load pattern (EagerInit on T:04 + main-thread load on T:01) is expected and reflects the overlapping parallel load architecture; the profiler records both independently. DB Load wall time is longer than v2.6.2 (684ms vs 371ms EagerInit) because both paths run to completion before the profiler deducts overlap; effective TTI impact is the longer of the two (~502ms main-thread path).

USER GUIDE VERSION HISTORY

Date	Version	Document
Sunday, December 17, 2006	Rev. a	Concept Doc
Tuesday, June 26, 2007	Rev. b	Design Doc
Saturday, April 09, 2011	Rev. c	Design Doc
Sunday, May 26, 2011	1.0	User Guide

Sunday, June 12, 2011	1.5	User Guide
Sunday, October 16, 2011	1.5.3	User Guide
Wednesday, November 02, 2011	1.5.5	User Guide
Sunday, November 20, 2016	1.7	User Guide
Tuesday, January 15, 2019	1.8.1.2	User Guide
Saturday, April 17, 2021	1.8.4	User Guide
Friday, May 5, 2023	1.8.5	User Guide
Thursday, September 30, 2025	2.1.1	User Guide
Wednesday, October 15, 2025	2.2.2	User Guide
Thursday, November 27, 2025	2.4.7	User Guide
Saturday, December 6, 2025	2.4.9	User Guide
Friday, December 26, 2025	2.4.10	User Guide
Monday, February 23, 2026	2.4.14	User Guide
Wednesday, March 25, 2026	2.6.2	User Guide
Sunday, May 24, 2026	2.8.4	User Guide

CREDITS

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